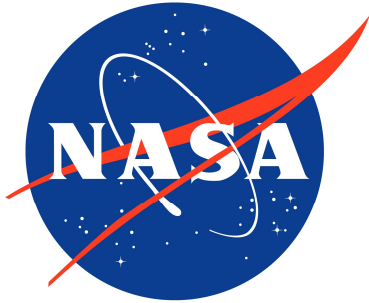


NASA/L'SPACE-2024-T24



System Requirements Review

Understanding the character and origin of lunar pole volatiles

Ice Hunters, Team 24

National Aeronautics and
Space Administration

NASA Lucy Student Pipeline
Accelerator and Competency Enabler

2024

Table of Contents

Table of Figures	2
Table of Acronyms	3
Ice Hunter SRR Outline	4
1. System Requirements Review	4
1.1. Mission Statement	4
1.2. Science Traceability Matrix	5
1.3. Summary of Mission Location	6
1.4. Mission Requirements	9
1.5. System Definition	11
1.5.1. Spacecraft Overview	11
1.5.2. Mechanical Subsystem	15
1.5.2.1. Mechanical Subsystem Requirements	16
1.5.2.2. Mechanical Subsystem Overview	17
1.5.2.3. Mechanical Subsystem Trade Studies	20
1.5.3. Power Subsystem	26
1.5.3.1. Power Subsystem Requirements	26
1.5.3.2. Power Subsystem Overview	27
1.5.3.3. Power Subsystem Trade Studies	32
1.5.4. Command and Data Handling (CDH) Subsystem	38
1.5.4.1. CDH Subsystem Requirements	38
1.5.4.2. CDH Subsystem Overview	40
1.5.4.3. CDH Subsystem Trade Studies	45
1.5.5. Thermal Management Subsystem	50
1.5.5.1. Thermal Management Subsystem Requirements	50
1.5.5.2. Thermal Management Subsystem Overview	52
1.5.5.3. Thermal Management Subsystem Trade Studies	54
1.5.6 Payload Subsystem	57
1.5.6.1 Payload Subsystem Requirements	58
1.5.6.2 Payload Subsystem Overview	59
1.5.6.3 Payload Subsystem Trade Studies	60
1.5.7 Recovery and Redundancy	62
1.5.8 Interface Control	63
1.6. Risk Analysis	64
1.7. Programmatics	67
1.7.1. Team Organization	67
1.7.2. Cost Estimate	67
1.7.3. Schedule Estimate	71
1.7.4 Change Control	74
1.8. Conclusion	74

Table of Figures

Figure 1. JMARS Haworth region (25 km reference).....	7
Figure 2. JMARS Haworth candidate 1 (500 m reference).....	7
Figure 3. JMARS Haworth candidate 2 (500 m reference).....	8
Figure 4. JMARS Haworth region (10 km reference).....	8
Figure 5. Engineering drawing (Wheel).....	14
Figure 6. Engineering drawing (Suspension).....	15
Figure 7. Engineering drawing (Suspension cont.).....	16
Figure 8. Engineering drawing (Chassis).....	17
Figure 9. Mechanical components specifications.....	22
Figure 10. Power distribution map.....	33
Figure 11. Power components specifications.....	34
Figure 12. CDH components specifications.....	46
Figure 13. Thermal components specifications.....	58
Figure 14. Team structure/roles.....	72

Table of Acronyms

CME	Coronal Mass Ejection
EDS	Electrodynamic Dust Shield
ISRU	In-Situ Resource Utilization
ESDMD	Exploration Systems Development Mission Directorate
NASA	National Aeronautics and Space Administration
LRO	Lunar Reconnaissance Orbiter
PSR	Permanently Shadowed Region
NIRVSS	Near-Infrared Volatiles Spectrometer System
NSS	Neutron Spectrometer System
TBD	To Be Determined
TRL	Technology Readiness Level
LRO	Lunar Reconnaissance Orbiter
CPU	Central Processing Unit
LEO	Low Earth Orbit
CIGS	Copper Indium Gallium Selenide
IMM	Metamorphic Multi-Junction
RTG	Radioisotope Thermoelectric Generator

Ice Hunter SRR Outline

1. System Requirements Review

1.1. Mission Statement

The Ice Hunter lunar rover mission concept, supported by NASA's Exploration Systems Development Mission Directorate (ESDMD) and aligned with the Artemis Accords, aims to conduct essential surface measurements and mapping to advance understanding of lunar water distribution and abundance. The mission will explore the PSR of the lunar poles. This exploration is to further the understanding of the changes in volatile materials. Furthermore, the mission will further the understanding of the compositional state and distribution of the volatile material it encounters. These goals align with the mission objectives, which are to collect 100g of ice samples from our chosen location, determine the water ice abundance in or near a PSR within the top one meter of regolith, and to identify areas of higher ice concentrations with an Artemis III landing site. The chosen location, Haworth, was chosen for its PSR in a crater showing promise of ice in previous reconnaissance missions, its low slope, and intermediate sunlight among other benefits. The two scientific instruments, NSS and NIRVSS, allow the ability for the rover to achieve the mission objectives through the instruments ability to detect hydrogen within the PSR, and then drill for samples within the PSR that the instruments can then more finely analyze. This mission plans to operate on the lunar surface for a full Earth year, consistent with the Artemis mission framework.

1.2. Science Traceability Matrix

The following table will provide the link between the science goals driving the mission and the requirements that will be set by the client, these two contributing factors will start to root the instruments that will be used to conduct the science experiments to be performed during the life of the mission. The instrumentation selected will allow the completion of the science goals by allowing for the detection and analysis of samples from the PSR found on the lunar poles. The goals centered around understanding the composition of the PSR will allow for a plan defined by objectives, including collecting samples, checking if the samples contain H₂O, and a location for which the rover can achieve these goals. Certain specifications and parameters will set the standards for the science to be performed and are specified in the table below:

Science Goals	Science	Science	Instrument	Predicted	Instrument	Mission
---------------	---------	---------	------------	-----------	------------	---------

	Objectives	Measurement Requirements		Performance Requirements		d Instrument Performance		Requirements
		Physical Parameters	Observables					
Understanding the character and origin of lunar poles (Primary) Understand the transport, retention, and alteration, and loss processes that operate on volatile materials near and at permanently shadowed regions (NASA, “Artemis III Science.”) (Secondary) Determine the compositional state (elemental, isotopic, mineralogic) and compositional distribution (lateral and with depth) of the volatile component in lunar polar regions (NASA, “Artemis III Science.”)	Collect 100g of ice samples at one of the 13 Artemis III landing site locations and determine the composition of the sample	Identify the concentration of Hydrogen wavelengths in a collected sample of lunar regolith	Detect a difference in water concentration from collected samples versus ground samples to predict loss of volatile material	Wavelength range:	250-600 nm	230-615 nm	Neutron Spectrometer System (NSS)	Vehicle shall traverse to PSR to scan/drill science locations
				Integration time	1 s	.5 s		
				Sensitivity:	10-2 to below 10-4 w/w	10-2 to below 10-4 w/w		
				Spatial resolution:	<100 μm	<100 μm		Mission must collect a minimum sample of 100 g of lunar regolith from a depth up to 1 m
	Determine the water ice abundance of at least one of the 13 Artemis III landing site locations in or near a PSR within the top 1 meter of regolith	Determine the percent concentration of Hydrogen within the top 1 meter of regolith	Detect Hydrogen wavelengths in the range of 400-680 nm	Wavelength range:	250-600 nm	200-610 nm	Near Infrared Volatiles Spectrometer System (NIRVSS)	
				Integration time:	1 s	1.2 s		
				Sensitivity:	10-2 to below 10-4 w/w	10-1 to below 10-5 w/w		
				Spatial resolution:	30 μm	15 μm		
	Identify areas of higher ice concentration within an Artemis III landing site	Compare concentrations of hydrogen on a 10 x 10 grid of 1 m^2 squares and	Detect areas of high ice concentration to predict ice collection sites for future	Wavelength range:	400-700 nm	350-750 nm	NSS & NIRVSS	Mission must scan lunar surface using the two science instruments
				Spatial resolution:	20 nm	10 nm		

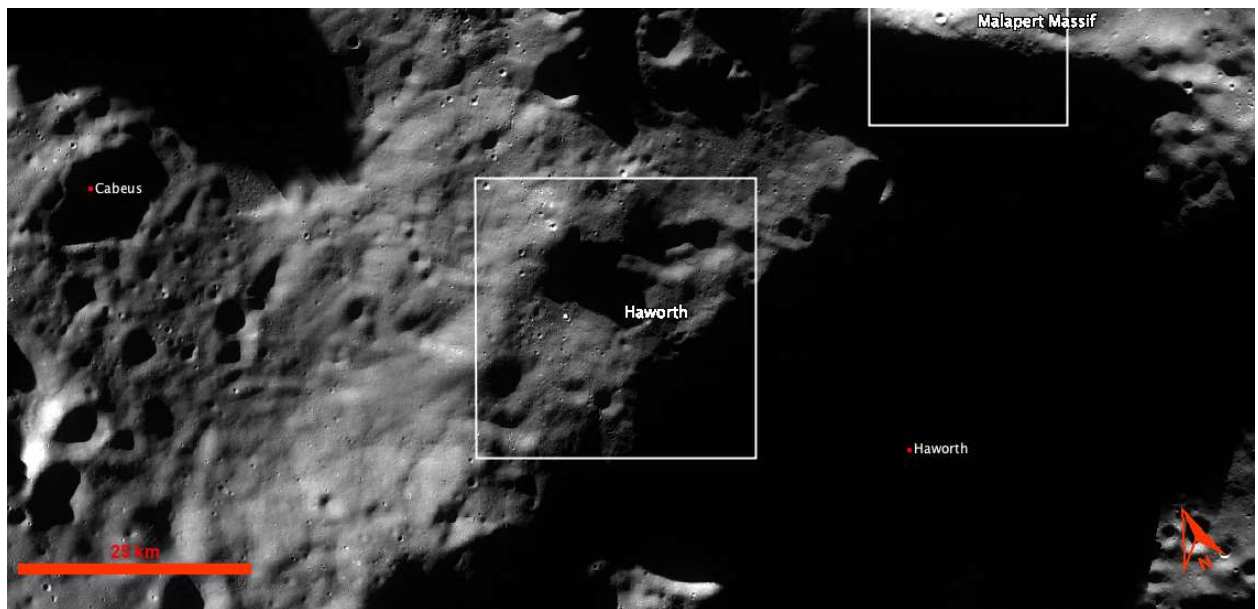
		identify 3 squares with the highest concentration of hydrogen	missions					
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1.3. Summary of Mission Location

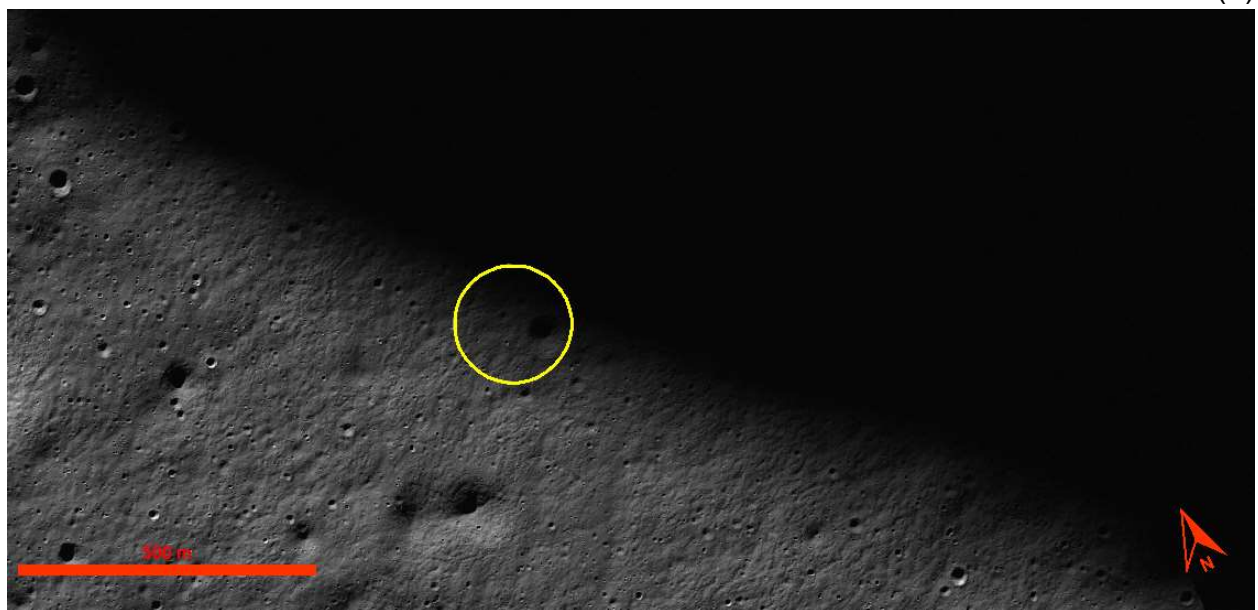
Upon reviewing identified candidate regions provided by the Artemis III landing sites catalog, the final decided location for the mission will be the Haworth region. This location has shown promise of potential water ice in the moon’s PSRs as previous reconnaissance missions have performed experiments to make these predictions such as the Clementine spacecraft (Clementine, 2023), the Lunar Prospector (Lunar Prospector, 2024), and the Lunar reconnaissance orbiter (LRO, 2024). All three missions have found evidence of ice in the bottom of PSR craters at the Moon’s south pole. The location selected has shown to be an optimal landing location due to its exposure to intermittent sunlight near the crater’s rims (Heldmann et al, 13). This intermittent sunlight will ensure the successful operation of the spacecraft as all relevant subsystems require the capturing of energy from the sun’s light. The coordinates for the location selected are -86.721°N, 338.283°E.

Intermittent sunlight along areas of the selected location will allow the spacecraft to enter PSRs, then exit science marked locations to recharge and regulate temperature as needed (Heldmann et al, 19). The science experiments can be completed within a reasonable range of distance between the PSR and areas of intermittent sunlight without high concern for the spacecraft’s charge. When entering and/or exiting a PSR, additional research will be completed in order to determine when the spacecraft is entering a shadowed region (not to be confused with a PSR) in order to ensure sufficient charge prior to planned operation cycles (Heldmann et al, 20).

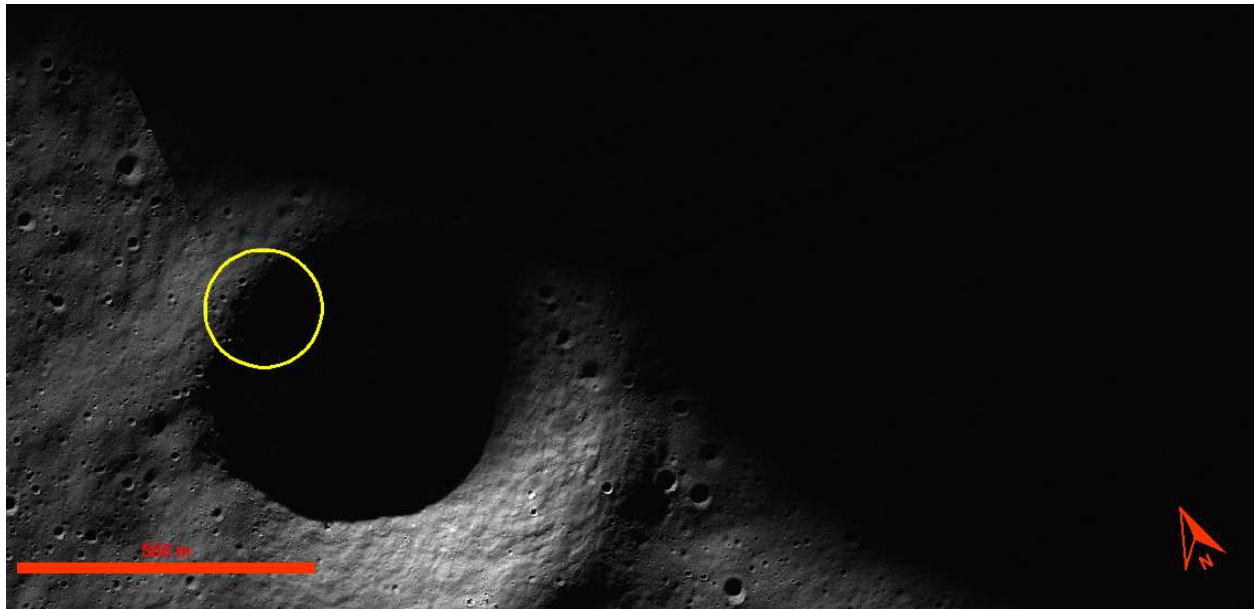
An additional benefit to the location selected is the multiple regions of flat terrain as constrained by the customer, each location highlight has met the requirement of having a slope <= 15°, Generally, the crater region offers a manageable slope for the craft to navigate and safely carry out science experiments. It also offers proximity to areas with higher elevation resulting in better and faster communication with the LRO (Jia et al). The following images were generated using the JMARS Luna software which displays the region of interest, along with specified locations for potential science experiments. Along these designated locations it can also be seen that sunlight won’t be far from these PSRs.



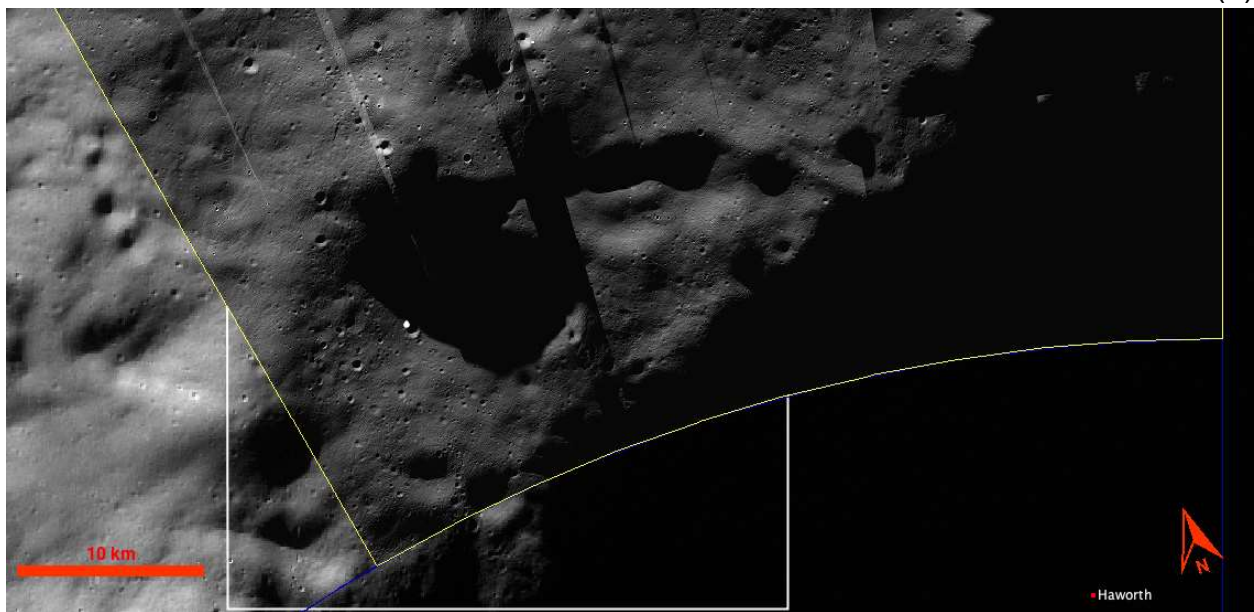
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1.4. Mission Requirements

As a secondary payload for a discovery class mission, the constraints given by the customer for the Ice Hunter mission are primarily cost, mass and volume. The cost allotted for the Ice Hunter mission shall not exceed \$225 million, which will include materials, science instruments, personnel costs and all related services, but will not include launch or cruise costs. The mass allotted for the Ice Hunter mission shall not exceed 85kg, including vehicle, science instruments, and all systems and subsystems required to accomplish the stated science goals. Finally, the total volume for the Ice Hunter mission shall not exceed a volume Length x Width x Height of 1.5m x 1.5m x

1.5m, though this is not fixed volume measurements, as the vehicle is permitted to expand and exceed these volume constraints post-deployment. Top level requirements for the mission are specified in the table below:

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
0.1	The system shall survive the lunar environment for TBD years	In order to achieve science objectives the system needs to operate/survive	Client	SYS-03	Demonstration	All	Met
0.2	Shall investigate the abundance of near surface water ice in PSR	Foundational science goal	Client	SYS-01,SYS-03,SYS-05,SYS-06,SYS-07,SYS-09	Test	Payload	Met
SYS-01	The system shall traverse the lunar surface by means of a rover	The rover must be able to travel distances for conducting science objectives	0.1,0.2	SYS-07,SYS-09	Demonstration	Mechanical, Electrical, GNC, Computer hardware	Met
SYS-02	The system shall not exceed a mass of 85 kg	Defined by launch provider	Launch provider	All	Inspection	All	Met
SYS-03	The system shall be able to operate at temperatures ranging from 90.15 K to 379.15 K	The rover must function in extreme temperatures of the lunar pole	0.1	SYS-07	Demonstration	Electrical, Thermal, Computer hardware	Met
SYS-04	The system shall not exceed a volume of L	Defined by launch	Launch provider	All	Inspection	Structural	Met

	1.5m x W 1.5m x H 1.5m	provider					
SYS-05	The system shall collect and store data from the two science instruments on the rover	The data collected will contribute to achieving the science goal	0.1,0.2	SYS-01,SYS-03,SYS-07	Demonstration	Science, Electrical	Met
SYS-06	The system shall communicate with the LRO	After data has been collected and analyzed it will be relayed to earth	0.1,0.2	SYS-07	Demonstration	Electrical, Computer hardware, GNC	Met
SYS-07	The system shall have sufficient power to survive/operate	The rover requires power to perform and communicate all objectives	0.1,0.2	SYS-03	Demonstration	Electrical	Met
SYS-08	The system shall not exceed a cost of \$225M	Defined by contracting client	Contract provider	All	Verification	All	Met
SYS-09	The system shall contain the necessary instruments to support the collection from the scientific instruments	The rover needs the ability to run experiments on the Moon's surface	0.2	SYS-05,SYS-07	Demonstration	Science, Electrical, Computer hardware, Thermal	Met
SYS-10	The system shall not contain a Radioisotope Thermoelectric Generator(RTG)	The rover cannot have any derivative of this concept	Contract provider	All	Verification	All	Met
SYS-11	The system shall be ready for launch by	In order to align	Contract provider	All	Verification	All	Met

	September 1st, 2028	with launch provider's secondary payload					
SYS-12	The system shall not contain radioactive material exceeding a mass of 5g	Limited radioactive materials	Contract provider	All	Verification	All	Met

1.5. System Definition

1.5.1. Spacecraft Overview

The spacecraft will be the culmination of multiple subsystems working as one cohesive unit in order to operate and perform effectively and efficiently throughout the life of the mission. The communication and order of the subsystems shall be integrated to be practical and optimal for all of the spacecraft's operations. The top level requirements for the spacecraft are specified in the table below:

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
SS-01	The spacecraft system shall survive the extreme temperatures of the lunar pole	The spacecraft's survival is crucial in achieving science goal	0.1	ME-03, PS-01, PS-02, CDH-01, TCS-1	Test	All	Met
SS-02	The spacecraft system shall traverse slopes $\leq 15^\circ$	Traversing the location of interest requires a capable system	SYS-01, SYS-03	ME-06	Test	Electrical, CDH, Computer hardware	Met
SS-03	The spacecraft system shall not exceed a distance > 10 km from the landing zone	Decreasing distance traveled significantly reduces risks	SYS-01, SYS-03	N/A	Test	All	Met
SS-04	The spacecraft system	Collecting	SYS-07	ME-04	Test	All	Met

	shall be able to collect and store solar energy to power subsystems	energy for powering all subsystems is crucial					
SS-05	The spacecraft shall be able to use science payloads to collect data	Collecting data from the two science instruments are important in achieving science goal	0.1,0.2,SYS-01,SYS-03,SYS-05,	TBD	Test	All	Met
SS-06	The spacecraft system shall be able to communicate with the LRO	Relaying images and data will allow engineers to guide the craft and scientists to analyze data	SYS-06	TBD	Test	Electrical,CDH, Computer hardware	Met
SS-07	The spacecraft shall have components on board such as cameras and lights to assess and maneuver between science locations	Having reference photos and sufficient lighting will help guide the craft in the PSR	SYS-01	TBD	Test	Electrical,CDH, Computer hardware	Met

The spacecraft to be designed is a mobile vehicle in the form of a rover, this was decided as it is of interest to perform the intended science from multiple locations in order to get the most data possible. Many items must be considered when designing the spacecraft, initially the envelope of volume and mass will have a great impact on the overall design of the vehicle as these requirements will yield a small to medium sized craft in terms of planetary rovers. Another driver for the design of the spacecraft will be the science to be performed, the science experiments planned will require a certain design so that the data can be efficiently retrieved. These items are the initial main drivers of the design and have been considered as the preliminary design for the vehicle

have been discussed. The rover will have a box-shaped body/chassis which will store all the other subsystems and science payloads, the main body will also have some extended sub-bodies that will extend carrying cameras to map/scan the landscape, the other will allow two solar arrays to extend to a height that exceeds the volume envelope when fully extended. Lights on the front of the body will also be part of the design when traversing in the PSR. The spacecraft will have 6 wheels to traverse on the lunar surface, the suspension type adopted is the rocker-bogie system which has shown consistent success in similar environments. The spacecraft will primarily be made of aluminum among other low mass materials. The mission will require a capable spacecraft for an unprecedented region of the moon, all things have been considered in the design of the vehicle such as temperatures, obstacles, intermittent sunlight areas, shadowed areas, slopes, and surface type.

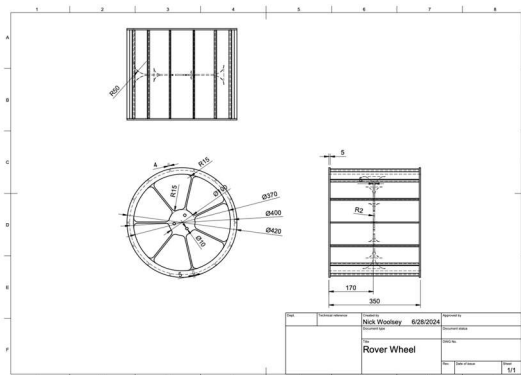
The mechanical subsystem of the spacecraft will provide mobility using the 6 wheeled rocker-bogie system allowing the spacecraft to traverse through the soft and rocky terrain of the lunar surface. The subsystem will primarily be made of lighter alloys with high strength that can perform in extreme temperatures. The subsystem will carry all other subsystems to survive and operate for the life of the mission. The subsystem will provide lighting for guidance in the PSR, it will also have an energy collecting system which will power the spacecraft and regulate temperature when in PSR. The mechanical subsystem will have two science instruments mounted to perform the science driving the mission.

The power subsystem generates, stores and distributes energy for all onboard systems that require power such as the CDH subsystem, Science payload and all Mechanical components like motors and actuators. This is done by utilizing the lightweight generation potential of solar panels to recharge the rover's batteries in between periods of testing inside the PSR of the crater. This energy is stored in batteries to power the rover while it is conducting this testing within the PSR and during transition periods between the charging point on the crater rim and the testing location within the crater. The distribution of this energy is done using a solar charge controller, an inverter and a power supply which effectively supplies the maximum amount of power required for the rover at all times.

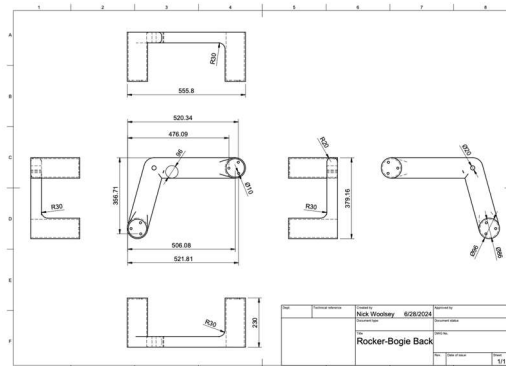
The Command and Data Handling subsystem is the brains of the spacecraft that is being outlined in this report. This subsystem has several specifications that must be fulfilled by the commercial off the shelf (COTS) parts that are being selected. These specifications are found in section 1.5.5.1 of this report and include; survivability, communication with the rover from Earth, data storage, internal craft communication, weight specifications, and power draw. The pieces of equipment that will uphold these requirements are the central processing unit (CPU), communication antenna, communication receiver, data bus, storage, and onboard software.

The thermal system will include multiple flexible heaters to keep the internal electronics warm in the frigid environment near and in the PSR. The radiator will be used to dissipate heat generated by the spacecraft to prevent overheating. The radiator will also prevent the heat of the sun from heating the rover above safe temperatures. As a backup, the a protective white thermal paint will be applied to the surface of the rover. A thermal switch will be installed to regulate between heating and cooling, depending on the temperature.

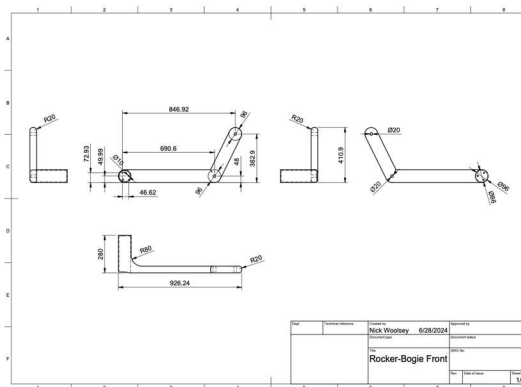
The science instrumentation to be used for this particular mission will survey and scan the surface of the moon using light to determine composition percentages that makeup these sample locations, the instruments used will consist of two high TRL devices including the NSS and NIRVSS instruments. These two scientific payloads will be important in achieving the science goal driving the mission.



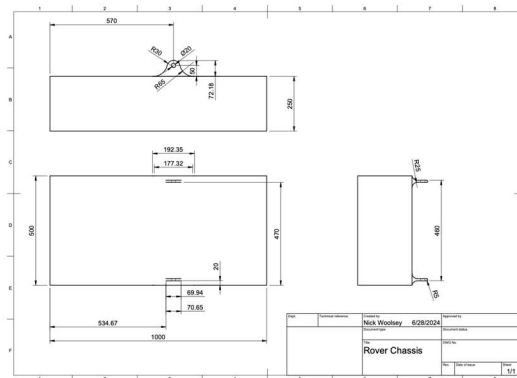
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1.5.2. Mechanical Subsystem

The mechanical subsystem will provide mobility and housing for all relevant subsystems and science payloads, the capabilities of the craft will be imperative in achieving science tasks/goal of the mission. The main items to consider in designing the subsystem are materials, mobility, and dimensions. The materials used for the craft must have a combination of low mass, high strength, and high performance under extreme

temperatures. A widely used material with these properties is aluminum specifically (2000,5000,6000 series) among other materials that will be considered in the assembly of the mechanical subsystem. Mobility is important when traversing the unprecedented terrain of the lunar south pole, a suspension system that can handle a rough and fine surface along with capable wheels such as mesh wheels or a rocker-bogie system will be desirable given their previous success in similar environments, other obstacles such as increasing slopes and lunar rocks are other situations to expect and design for. Finally the dimensions that make up the craft will be important in meeting the constraints of volume and mass by the client, as well as housing all the relevant subsystems to operate/maintain the mechanical subsystem, there are many components that go into the cumulative operation of the system and having enough space to store them all while providing protection from the harsh environment is key to the mission's success. Achieving the science goal will also help in driving the design of the subsystem as the science tasks may require the craft to be lower to the ground for drilling/scanning regolith. These foundational components of the mechanical subsystem have great importance and will ensure the success of the overall mission.

1.5.2.1. Mechanical Subsystem Requirements

The mechanical subsystem comprises the 4 main components, mobility, materials, suspension, and science collection systems. The successful design of these components to work together effectively will be a deciding factor in achieving the science goal driving the mission. Lower level requirements have been considered and evaluated, these requirements are specified in the table below:

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
ME-01	The subsystem shall operate/perform under extreme temperatures ranging from 90.15 K to 379.15 K	Achieving science goals relies on the successful operation of the mechanical subsystem	SYS-03	TBD	Test	Thermal	Met
ME-02	The subsystem shall use materials not exceeding a mass of 55 kgs	The mass of mechanical components will be limited	SYS-02	TBD	Inspection	All	Met

ME-03	The subsystem shall not exceed an allotted cost totaling 225 M of all subsystems combined	A finite amount of funds will be allotted for the mechanical subsystem	SYS-08	TBD	Inspection	All	Met
ME-04	The subsystem shall be able to collect energy by means of solar arrays/panels	In order to operate/perf orm the system needs sufficient power	SYS-07	TBD	Test	Electrical,CDH ,Thermal,Com puter hardware	Met
ME-05	The subsystem shall have sufficient volume to carry all payloads required to operate/perform tasks	The craft needs sufficient space to carry all necessary payloads	SYS-04	TBD	Inspection	All	Met
ME-06	The subsystem shall have a mobility system to overcome obstacles of the lunar surface such as rough/soft terrain	Science objectives/en ergy collection rely on the capabilities of the mobility system	SYS-01	TBD	Test	Electrical,CDH ,Computer hardware	Met
ME-07	The subsystem shall have a camera to scout paths of travel with the least obstacles	Cameras will help assess the location and investigate locations of interest	SYS-01	TBD	Test	Electrical,Com puter hardware	Met
ME-08	The subsystem shall have lights to guide the craft in the PSR	Lights will help navigate the craft in the dark environment	SYS-01	TBD	Test	Electrical	Met

1.5.2.2. Mechanical Subsystem Overview

In the process of developing the mechanical subsystem many factors contributed to the preliminary design, a combination of Technology Readiness Level (TRL) and practicality are of the highest priorities. The main components to be considered for the subsystem are suspension, materials, wheel design, and science collecting devices. Research of previous missions have shown how the adopted suspension type of the rocker-bogie system has been successful in similar environments providing a high TRL of 9, as this technology is flight proven through successful mission operations. Materials to be used in the construction of the subsystem will primarily be of aluminum, but also include titanium, carbon fiber, and plastics. Aluminum has proven to be very desired in space applications because of its great mass to strength ratio, many of the operating satellites, landers, and rovers have shown great success in this material, this will also provide a high TRL of 9 for primary materials used in the subsystem. The wheel design is imperative to the success of the mission, in order to collect science and power the vehicle must have capable wheels to traverse in the unprecedented lunar pole terrain, there have been many variations of redesigning the wheel for different space missions, the wheels will endure temperatures from 90.15 K in shaded regions to 379.15 K in sunlit areas. Single piece treaded wheels have shown to have fewer points of failure and although they may endure some wear damage, they still prove to be successful in their function. Past missions with this wheel type will provide a high TRL of 8, given that although successful, the previously tested environment doesn't apply to this particular terrain. The last component to consider will be a drill for collecting samples of regolith at different depths, a top candidate for this function is the trident drill, this drill can endure the surface of the moon as it cuts and hammers to fragment hard material (Rachel Hoover, NASA), this component will have the lowest TRL of all components at 6. All components having a high Technology Readiness level will contribute to the efficiency of cost, performance, and reliability, ensuring exponentially increased chances of a successful mission.

Suspension Type

The suspension system decided upon was a rocker-bogie style suspension. The main factor in this decision was the high level of heritage that this system has. For example, rocker-bogie suspensions are used in all of NASA's rovers including Curiosity, Spirit, Opportunity, and Perseverance with a high level of success.

Another key consideration was simplicity and manufacturability. We wanted a system that was as simple as possible to keep the amount of potential weak points to a minimum, keep costs down, and ensure the system could be manufactured in a timely manner. A rocker-bogie system offers all of these features. There are no springs or shocks to worry about, the design is simple and easy to implement, and it is reliable.

The simplicity of the system means fewer components that could malfunction and fewer mechanical parts that can get clogged with dust, fatigue, or wear. This is a huge benefit considering the rover will be hundreds of thousands of miles away where maintenance is not an option.

We also need to consider where we are going. The moon is a much different environment to Earth and with that come certain challenges to overcome. The moon has a sandy and dusty surface that could become an issue for traction and sinking. A rocker-bogie suspension has a natural ability to spread the load of the vehicle across all six wheels, which minimizes the maximum ground pressure on any one wheel. This is especially important when traversing soft terrain and ensures maximum traction with all six wheels on the ground at all times.

The combination of a rocker-bogie suspension and wheels as opposed to tracks completely negates the initial worry of overcoming obstacles. Generally, a wheel needs to be twice as tall as the object it is trying to overcome in order to go over it. With a rocker-bogie suspension involved this equation actually reverses and the rover will now be able to scale objects twice as tall as the wheels!

Material

It has been decided that our rover will use a combination of titanium, aluminum, carbon fiber, and plastics to optimize the design and functionality of the rover while keeping the weight to a minimum. Different materials will be used on different components of the rover depending on their needs and specifications. Initially, magnesium was considered as a construction material because of its incredible weight-saving potential, but its high reactivity posed an extreme risk to the mission and was consequently eliminated from consideration. Notable reactions include corrosion: magnesium is highly reactive in the presence of moisture, which is a primary concern for our mission. Flammability: magnesium is highly flammable and can burn intensely. This is a significant concern, particularly during the launch phase when materials are exposed to high temperatures. Special coatings and treatments are often used to mitigate this risk. While magnesium has good thermal conductivity, it also has a relatively low melting point compared to other metals like titanium or aluminum. This can be a disadvantage in certain high-temperature applications. Especially on the Moon where temperatures can reach 379.15 K.

Wheel design

Designing wheels for space applications can be challenging. Rovers on the moon will experience temperatures as low as 90.15 K which will cause traditional tires like what we use on Earth to shatter like glass. Because of this we need to design a wheel that can withstand those temperatures and remain durable.

Science collection

The mechanical subsystem will support the drill device that will contribute in achieving the science goal, this device will be integrated into the main body of the rover in order to dig at a depth of up to one meter of the lunar regolith. These drilling sessions at different sites will provide insightful data to the composition of the samples produced using this drill.

Component	Mass [kg]	Volume [m]	Max Power Draw [W]
Chassis	20	1 x .5 x .25	0
Suspension	5	.8 x .3 x .05	0
Wheels	15	.349	24
Drill	15	.206 x .333 x 1.68	3

(9)

1.5.2.3. Mechanical Subsystem Trade Studies

Tracks vs Wheels					
Criteria	Explanation	Grade	Weight	Wheels	Tracks
Reliability and Simplicity	A high level of reliability is needed since the rover will be on the moon and there won't be any engineers around to fix it if repair is needed,	10 = high, 5 = medium 1 = low 0 = Fail	25	10	2
Cost of Production	Ideally, the system should be as cost-efficient as possible.	10 = high, 5 = medium 1 = low 0 = Fail	10	9	4

Speed	Speed will determine how fast we are able to get from one location to another and could be important in collecting data in a reasonable amount of time.	10 = high, 5 = medium 1 = low 0 = Fail	10	8	2
Maneuverability	Terrain can be difficult to manage and having a high maneuverability to navigate the best path is ideal.	10 = high, 5 = medium 1 = low 0 = Fail	10	10	4
Lightweight	The rover will be on a rocket to get to the moon. Every pound needs to be accounted for and factored in, hence why we have a limit to our weight. In order to spread our weight requirement over all of our systems, we need a lightweight solution to movement.	10 = high, 5 = medium 1 = low 0 = Fail	20	9	3

Overcome Obstacles	How well can the system drive over obstacles and not get stuck? Very important in a rocky terrain where objects could be much taller than the rover.	10 = high, 5 = medium 1 = low 0 = Fail	10	6	10
Power Efficiency	We do not want a power-hungry system. The movement type should be as power-efficient as possible.	10 = high, 5 = medium 1 = low 0 = Fail	15	6	10
		TOTALS:	100%	85%	46%

A wheel has fewer moving parts, which means that there are fewer components that can get damaged than tracks. Wheels also generally have a lower cost of production than tracks. Due to more friction and a complex mechanical system, continuous tracks have lower speed compared with wheels. Compared with tracks, the wheels need a lower amount of torque to move the rover from a stationary position. Wheels provide higher maneuverability than continuous tracks. Continuous tracks are much heavier than wheels, and this is the main reason why wheels are used especially in cases when the mass of the robot is a critical property – for example, space exploration missions. Using continuous tracks, a UTV can operate on rough terrain where wheels would get stuck. Also, the continuous band of treads can ascend and descend stairs, surmount obstacles, or cross ditches. Depending on the terrain, a robot needs to pass small or large obstacles. For a wheel to get over a vertical obstacle, it has to be at least twice as tall as the vertical obstacle. Compared with wheels, continuous tracks have high performance and an optimized traction system, which is an advantage in power delivery efficiency.

Mesh <https://inhabitat.com/nasas-new-airless-titanium-tires-are-almost-indestructible/>

https://link.springer.com/chapter/10.1007/978-3-030-89784-0_16

<https://metalprofy.com/versus/titanium-vs-aluminum/#:~:text=Titanium%20also%20has%20excellent%20thermal%20stability%2C%20which%20means,weight%2C%20better%20corrosion%20resistance%20and%20higher%20temperature%20tolerance.>

Mesh Wheels vs Single-Piece Wheels					
Criteria	Explanation	Grade	Weight	Mesh	Single-Piece
Reliability and Simplicity	A high level of reliability is needed since the rover will be on the moon and there won't be any engineers around to fix it if repair is needed,	10 = high, 5 = medium 1 = low 0 = Fail	30	5	8
Cost of Production	Ideally, the system should be as cost-efficient as possible.	10 = high, 5 = medium 1 = low 0 = Fail	10	6	9
Heritage	Previous missions or research are tremendously valuable here to prove the wheel's capability. A high level of heritage is extremely preferred.	10 = high, 5 = medium 1 = low 0 = Fail	10	6	8

Lightweight	The rover will be on a rocket to get to the moon. Every pound needs to be accounted for and factored in, hence why we have a limit to our weight. In order to spread our weight requirement over all of our systems, we need a lightweight solution.	10 = high, 5 = medium 1 = low 0 = Fail	30	10	8
Overcome Obstacles	How well can the wheel drive over obstacles? Very important in a rocky terrain to extend the life of the rover.	10 = high, 5 = medium 1 = low 0 = Fail	20	8	7
		TOTALS:	100%	73%	79%

Mesh wheels are much more complex than single-piece wheels. Mesh wheels have a lot of parts and with that, more parts to inspect, more fasteners to install, and more things that can go wrong. On the other hand, the single-piece wheels are generally milled out of metal and there is very little that can go wrong or needs to be inspected. NASA's Curiosity rover is still running and functioning even though the wheels have shown noticeable wear. The cost of each wheel style is difficult to determine but it is likely that the mesh wheel would cost more with the added complexity and amount of parts. The most notable NASA mission featuring mesh tires is the Lunar Roving Vehicle and most recently NASA's new mesh wheels which have not been implemented or proven. Single-piece tires have been featured on several NASA rovers. Mesh tires are very weight efficient and will always beat out solid designs, making mesh wheels here the clear choice. Mesh wheels have an ability to deform and conform to the shape of landscape which has proven to be useful in the Lunar Roving Vehicle and higher speed applications. Solid single-piece wheels generally have a tread applied that can also aid in traversing terrain.

Skid Steering vs 4-Wheel Steering					
Criteria	Explanation	Grade	Weight	Skid Steering	4-Wheel Steering
Reliability and Simplicity	A high level of reliability is needed since the rover will be on the moon and there won't be any engineers around to fix it if repair is needed,	10 = high, 5 = medium 1 = low 0 = Fail	30	10	6
Cost of Production	Ideally, the system should be as cost-efficient as possible.	10 = high, 5 = medium 1 = low 0 = Fail	10	8	5
Heritage	Previous missions or research are tremendously valuable here to prove the wheel's capability. A high level of heritage is extremely preferred.	10 = high, 5 = medium 1 = low 0 = Fail	20	1	10

Lightweight	The rover will be on a rocket to get to the moon. Every pound needs to be accounted for and factored in, hence why we have a limit to our weight. In order to spread our weight requirement over all of our systems, we need a lightweight solution.	10 = high, 5 = medium 1 = low 0 = Fail	10	10	6
Overcome Obstacles	How well can the wheel drive over obstacles? Very important in a rocky terrain to extend the life of the rover.	10 = high, 5 = medium 1 = low 0 = Fail	10	8	10
Power Efficiency	We do not want a power-hungry system. The movement type should be as power-efficient as possible.	10 = high, 5 = medium 1 = low 0 = Fail	20	6	10
		TOTALS:	100%	70%	79%

Skid steering is compact and requires few parts meaning less points of failure and less things to go wrong. Skid steering keeps complexity to a minimum whereas the downside of 4-wheel steering is a higher actuator count, part count, and higher level of complexity. Because skid steering requires no additional parts for steering it is much more cost efficient than 4-wheel steering which requires many more parts to function well. Skid steering is rare and not featured on any of NASA's well-known rovers whereas a 4-wheel steering system is featured on many different rovers on many different missions. Again, with skid steering having fewer parts to function it comes out much lighter than a 4-wheel steering system with many additional parts. Skid steering does not have as aggressive of steering mechanics as 4-wheel steering. This means skid steering will be

inhibited before 4-wheel steering is inhibited because 4-wheel steering points the wheels in the direction of travel so that skidding is minimized. Skidding causes unpredictable power requirements. The advantage of 4-wheel steering is lower power consumption. 4-wheel steering draws significantly less power than skid steering during tight turning maneuvers.

1.5.3. Power Subsystem

The power subsystem must allow for the rover to perform the specific tasks outlined in the mission task document. To do this it must produce and store the required amount of power needed for the rover during charging periods outside of a PSR and while conducting data collection within a PSR. Due to the rover entering a crater's PSR, specifically the Haworth crater, the power subsystem needs to take into account the ability for the rover to regenerate enough power in order to recharge the onboard batteries and store enough power in order to enter the PSR, collect data, samples and performs tests and then exit the crater's PSR all under its own power to reach a suitable charging location.

1.5.3.1. Power Subsystem Requirements

The following table defines the requirements needed for the power subsystem. These requirements are based on the mission requirements, which outline the needs of the rover while in the defined location and in the PSR. Furthermore, by understanding the needs of the other subsystems and the payload instruments themselves, we get an understanding of the needs the power subsystem must fulfill.

Requirement Number	Requirement Description	Rationale	Parent Req	Child Req	Verification Method	Relevant Subsystem	Requirement met?
PS-01	The Power Subsystem shall be able to withstand the temperature range on the lunar surface. (90.15 K to 379.15 K)	The system must be able to work in the operating conditions on the moon.	SYS-03		Test	Thermal Control Subsystem	Yes
PS-02	The Power Subsystem shall be able to withstand the environmental radiation on the surface of the moon. (1,369 microsieverts)	The system must be able to function on the surface of the moon for the lifetime of the mission	PS-06 PS-07		Test and Specifications	Power subsystem	Untested

	per day)						
PS-03	The Power Subsystem shall be able to supply all power requirements of the spacecraft between solar charging intervals. (1 week hold)	The system must be able to ensure that the science goals of the craft can be fulfilled and therefore must be able to supply all instruments with power for up to 1 week	SYS-07		Test and calculations	Power generation and storage	Untested
PS-04	The Power Subsystem shall weigh no more than 20 percent of the total allotted kilograms. (17.1 kg)	The system must not encroach on the maximum weight requirement for the whole craft	SYS-02		Measurement	N/A	Up for Revision
PS-05	The Power Subsystem shall require no more volume than 20 percent of the total allotted craft volume. (0.675m ³)	The system must not encroach on the volume requirements of the spacecraft	SYS-04		Measurement	N/A	Yes
PS-06	The Power Subsystem shall be no less than 75 percent efficient.	The system must not restrict the operation of the craft excessively or to a point where it draws more power than the rest of the craft.	SYS-07	PS-03	Test	Thermal Control Subsystem	Untested
PS-07	The power subsystem shall be able to generate and supply 275 watts at peak rover operation.	The power subsystem must be able to supply the craft with power at peak operation	N/A	PS-03	Calculation and test	Computer Design Hardware, Mechanical Subsystem, Science Payload Subsystem	Yes

1.5.3.2. Power Subsystem Overview

In this section, the power system of the rover and its design will be explained in detail to discuss the component choices, their advantages and their disadvantages in order to sufficiently supply the craft with power at peak loads and for extended periods of time of up to 1 week in permanently shadowed regions. By doing this this section hopes to ascertain the capabilities of the craft and discuss steps to mitigate any foreseen issues or inherent disadvantages within the power subsystem. The first step in this process is acquiring information about the power usage of the craft so we can effectively design a system that is sufficiently lightweight, powerful and reliable to fulfill the requirements of the craft and subsystem outlined in section 1.4 of the MCR and section 1.5.4.1 of this document.

We will begin this process by looking at the power draw first for the mechanical components on this craft, then the computer design hardware components, and finally the science payload. For the mechanical components on this craft there are three separate types of components that draw power, the hub motors of which there are six Maxon EC 32 flat motors, linear actuators of which there are two LINAK LA-12 linear actuators, and rotary actuators or servos of which there are three BETU 45kg Coreless Servo Motors. The electrical rating of the hub motors is 42.6mA at 24V for each motor, resulting in an individual power draw of 1.1W per motor. Thus, for all motors there is a total possible current draw of 255.6mA and power draw of 6.6W. For the linear actuators the power draw is 1A and 24V under full drain each, this gives us a power draw of 24W each for a total of 48W. Last, for the rotary actuators and servo's, each has a power draw of 4A and 8.3V which when calculated yields a power draw of 16.8W each for a total of 50.4W when all three servo's are combined. The total power draw for the mechanical components in this section is approximately 103.7W at peak power draw assuming all components are under load and operating at the same time.

Moving on to the computer design hardware, there are several components that have notable power draws. First is the heart of the CDH sub system, the motherboard/CPU in the CDH subsystem draws approximately 6.7A and 5V adding up to a little under 35W at maximum operation, one important note for the power subsystem is that the component will likely never experience this amount of power draw unless transmitting or receiving information nonetheless it will be assumed that the power draw is constant for the sake of margin and unforeseen computational loads. Next, both the transmitter and receiver array draw approximately 55W each for a total of 110W. For the memory that the CDH system will use the motherboard/CPU has allocated space and power draw for this part but it is worth calculating as if it were by itself to ensure that it does not impact the function of the power system. This component draws very little power with only a 30mA and 3.3V draw totaling to approximately 0.12W in read/write mode for 8MB of nand flash memory. Knowing this we can assume that the power draw while in

read/write mode will not exceed the allotted power draw that the motherboard/CPU single board computer has allotted for in its own power draw.

Finally, moving onto the last major system on the spacecraft, the science payload has two major pieces of equipment that need to be powered. First is the NIRVSS which uses 29.6W of power when in use and second is the NSS which uses 1.6W of power when in use. These two parts use a combined 31.2W of power for the whole science payload subsystem. All in all the craft uses approximately 274.9W when every piece of equipment is in full operation. This information will be used to ascertain how many batteries are needed to power the craft in between charging intervals on the crater rim. For these necessary charging intervals the craft will need to utilize solar cells in order to generate energy, as solar cells allow the rover to carry more power while taking up less weight than if it were to carry all of the necessary energy in batteries for the length of the mission.

The solar cells that the power subsystem team has chosen for this mission are the ZTJ Space Solar Cell from Rocket labs. This solar cell was chosen because it provides excellent low light and low temperature performance as well as a flight proven and unique tri fold storage that saves volume. The rover that is being designed by this team will have approximately 165 square centimeters or 0.0165 square meters of solar panel surface area exposed to the light when unfolded. This will provide the power subsystem of the craft with approximately 400W of peak power—enough to charge the batteries and operate the rover at the same time. Overall, this is a good choice because of its high Technology Readiness Level (TRL) and its peak output of 400W at 37V and 14A which when flowing does not overpower the AIMS Power 30 AMP Solar Charge Controller which has a peak operating limit of 800W. This is important because the solar power controller exists to manage and direct the flow of power between the solar panels and the battery, as well as from the battery to the rest of the system. The conjunction of the solar panel and solar power controller should provide the necessary energy to charge the rover's batteries that will be used to supply the craft with power when not in direct sunlight.

The rover will likely use 7 Eaglepicher 55 Ah lithium ion battery cells. These cells offer several advantages over other cells like nickel hydrogen which is commonly used in space such as a higher power density of over 340 Wh/L, a larger operating temperature over some models of nickel hydrogen which allows us to operate in colder and warmer environments without relying on the thermal subsystem. On top of this these cells are lightweight at under two kilograms each and have a slightly higher voltage output compared to other models which allows the battery pack to contain less parts assembled on site reducing the risk of failure. And lastly they contain the necessary voltage and current for a 24VDC to 120VAC jump when the stored energy passes

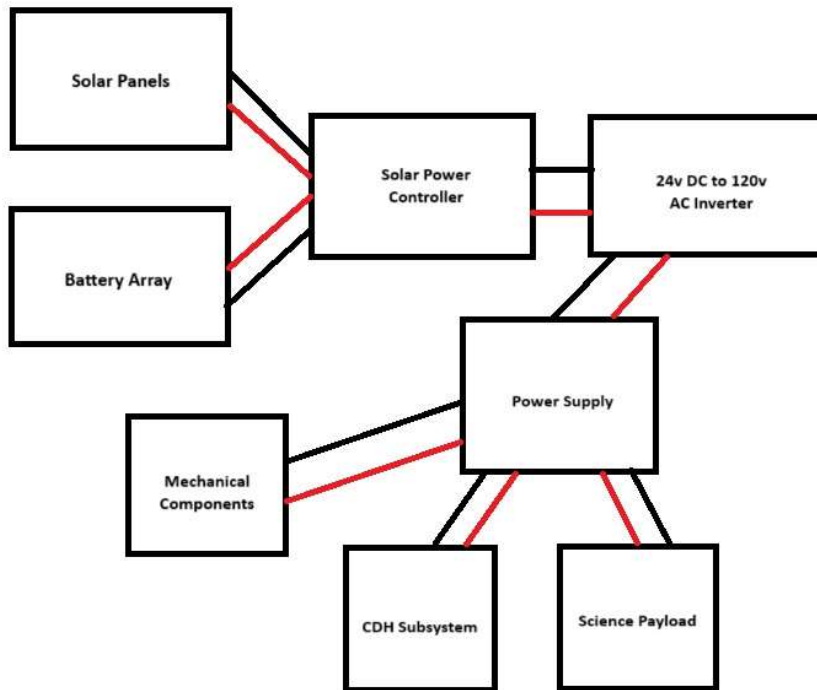
through an DC to AC inverter to bring the voltage and type of current up to the required specifications of the power supply.

The DC to AC inverter that we will be using is a 24V 300W Inverter, 24V to 120V made by and sourced from inverter.com. This inverter specifically fits several of our needs. Firstly it has a normal operating level of 300W which is just above our total power draw for the rover which allows the power system to fulfill requirements PS-03 and SYS-07. On top of this, the thermal operating range for the inverter is between -20 and 50 degrees celsius. This is similar to other components in the power subsystem and will therefore require similar thermal control.

Following the flow of power from the inverter to the power supply, the power subteam selected the Mean Well SPV-300 series AC/DC Power supply. This power supply was selected because of its high thermal range of -25 to 60 degrees celsius which matches other thermal requirements for the power subsystem and More importantly this power supply has an operating load of 300W which is just above the power requirements for the rover. On top of this, the power supply has a trimmable voltage and current for each output of up to 26V and 12.5A from the standard output voltage and current. This allows the power subsystem team to cater to the needs of each subsystem and supply low voltage parts like the CPU/motherboard in the CDH subsystem with power.

This concludes the main spine of the power subsystem. Summing up this information, the components in this system are; one, 1.65 square meter ZTJ Space Solar Cell, one battery array consisting of six batteries joined in series, one Solar power controller rated to 800W, one 24VDC to 120VAC inverter, and finally one 24VDC power supply. The next section of this report will discuss the technology readiness levels (TRL's), total

weight of the power subsystem and the max power draw for all subassemblies.



(10)

This sub assembly was constructed with cost in mind. While some of the components like the batteries and Solar panels have high TRL's in the 7 through 9 range most of the systems components are not flight proven. This poses an issue as while the readily available commercial products bring down the cost of the system, the TRL for the system as a whole is defined by the Solar Charge Controller which has a TRL of 5 as it had been proven commercially in both heat and cold but remains untested outside of commercial use. This stands for the power supply and DC to AC Inverter as well meaning that the system has a TRL of 5 when counting all pieces. Nonetheless these components should not be drastically affected by the space environment provided they are properly thermally managed because they do not use software or drivers that can become corrupted in a high radiation environment. Therefore to ensure that these components work in the lunar environment these components will need testing in a simulated space environment and for vibration resilience to ensure they do not break. This will raise the TRL of these components to a TRL of 7 when tested with all of the power system components. The next part of the power system outline goes over the graph below which includes weight and volume, max power draw for all of the components and power capability for the power subsystem to ensure that the power system will align with the requirements of the rover.

Table 0.0: Power Subsystem Parts and their weight (kg), volume (cm), power handling requirements (watts), and power capabilities (watts)

Part Name	Weight (kilograms)	Volume (centimeters)	Power Handling Requirements (watts)	Power Capabilities (watts)
Solar Panels	1.4 kg	13 x 13 x 1 cm	Supply 400W	400W
Solar Charge Converter	2.3 kg	20 x 17.5 x 7 cm	400W	800W
Battery Array	11.8 kg	96 x 98 x 23.8 cm	Supply 300W	310W
DC to AC Converter	0.730 kg	19.5 x 11.5 x 6.0 cm	275W	300W
Power Supply	1.1 kg	21.5 x 11.5 x 5.0 cm	275W	300W
Total:	17.33 kg	226288.75 cm ³ (0.6m ³)		

(11)

To conclude this section the percentages of weight and volume for this craft in terms of the allotted space will be displayed and discussed. For the total weight of the craft the power subsystem makes up about 20 percent of the total allotted weight. The power subsystem team believes that this is reasonable considering the heavy components like batteries that are required for this system to function. If necessary however, there are several components that could be changed to lower this net weight such as the batteries and power supply. Carrying on, the total volume of the subsystem is approximately 0.6m³ which makes up approximately 18 percent of the 3.375 meter required maximum volume for the spacecraft. This can also be lowered by decreasing the size of the battery array or making other components in the power subsystem smaller. Overall the capacity of the power subsystem when taken into account with the weight and volume requirements along with the systems TRL after testing make the power systems subteam confident in the viability of this design.

1.5.3.3.

1.5.3.4. Power Subsystem Trade Studies

Solar Cell

Criteria	Explanation	Grade	Weight	CTJ30	IMM	CIGS
Cost	A lower cost will allow for more funds to be allocated to other subcomponents and subsystems	10 = low cost, 5 = 1 = high cost	30%	6	8	7
Efficiency	A higher efficiency will decrease the necessary size of the panel and provide reliable power to other subsystems	10 = high efficiency rating, 5 = average efficiency rating 1 = poor efficiency rating	40%	8	9	6
Mass	A smaller mass will allow more mass to be allocated towards other subsystems	10 = light, 5 = medium 1 = heavy	30%	8	9	8
		TOTALS:	100%	74%	87%	69%

CIGS <https://scitechdaily.com/23-64-efficiency-scientists-set-new-world-record-for-cigs-solar-cells/>

IMM https://www.nasa.gov/wp-content/uploads/2021/10/3.soa_power_2021.pdf

CTJ30

<https://connectivity.esa.int/projects/ctj30class>

Electronic Control Unit / Power Supply Trade Study						
Criteria	Explanation	Grade	Weight	Mean Well SPV-300 series AC - DC Power	Keithley 2230-30-1 Program mable DC	Bravo EPP-300-24 Dc power supply

				supply	Power Supply	
Energy	Must be able to handle the power load required by the system	10 = supply over 275w 5 = supply over 250w 1 = low supply 0 = Fail	20%	10	6	7
Temperature	ECU must be able to operate from 223.15 K to 323.15 K	10 = within temp range 5 = close to temp range 1 = just outside temp range 0 = Fail	20%	7	6	10
Voltage	Must be able to supply 24V at maximum power from the batteries	10 = supplies 24v 5 = supplies close to 24v 1 = no 24v but trimmable output 0 = Fail	20%	10	7	5
Weight	ECU must weigh less than xxxx kilograms	10 = weight < 1kg 5 = weight < 5kg 1 = weight < 10kg 0 = Fail	20%	7	1	10
Power Type	DC to DC power supply (does not require AC to DC converter)	10 = Pass 0 = Fail	20%	0	0	0
		TOTALS:	100%	70,00%	42,00%	66,00%

[Rent or Buy Keithley 2230-30-1 Programmable DC Power Supply, Triple, 6V, 5A / \(2x\) 30V, 1.5A \(axiomtest.com\)](http://www.axiomtest.com)

<https://www.meanwellusa.com/upload/pdf/spv-300/spv-300-spec.pdf>

<https://www.bravoelectro.com/amfile/file/download/file/5471/product/7223/>

Solar Controller Trade Study						
Criteria	Explanation	Grade	Weight	AIMS Power 30 AMP Solar Charge Controller 12/24 VDC	AIMS Power 10 AMP Solar Charge Controller PWM 12/24 VDC	Charge C
Energy	Must be able to take an input from the solar panels and charge the batteries and or power the rover	10 = amps > 30 5 = amps > 20 1 = amps > 10 0 = Fail	25%	8	6	8
Temperature	must be able to function from 223.15 K to 323.15 K	10 = full function in the temp range 5 = low function in the temp range 1 = little function 0 = Fail	25%	9	9	8
Voltage	must be able to handle 24V or greater voltage potentials	10 = up to 48V 5 = 24V and up 1 = near 24V 0 = Fail	25%	7	6	10

Weight	The solar controller must be lightweight	10 = weight < 1kg 5 = weight < 2kg 1 = weight < 3kg 0 = Fail	25%	7	10	4
		TOTALS:	100%	77,50%	77,50%	75,00%

[AIMS Power 30 AMP Solar Charge Controller 12/24 VDC MPPT, SCC30AMPPT \(globalindustrial.com\)](#)

[AIMS Power 10 AMP Solar Charge Controller PWM 12/24 VDC, SCC10APWM \(globalindustrial.com\)](#)

[AIMS Power 40 AMP Solar Charge Controller 12/24/36/48 VDC MPPT, SCC40AMPPT \(globalindustrial.com\)](#)

Battery Type Trade Study						
Criteria	Explanation	Grade	Weight	Nickel Cadmium Batteries	Nickel Hydrogen Batteries	Lithium ion
Energy Density	Battery must be able to store energy compactly and in a weight efficient manner	10 = <500 wh/kg 5 = < 150 wh/kg 1 = < 50 wh/kg 0 = Fail	25%	1	5	10
Temperature	Battery pack must be able to provide power at low temperature ranges	10 = -50 to 50 C 5 = -25 to 50 C 1 = 0 to 50 C 0 = Fail	25%	7	10	5

Reliability	Are the batteries reliable and how long does it take for them to fail? What happens when they fail?	10 = over 10000 cycles 5 = over 1000 cycles 1 = over 100 cycles 0 = Fail	25%	7	10	5
Discharge Voltage	Discharge rate and voltage output over battery drain.	10 = low voltage change 5 = medium voltage change 1 = high voltage change 0 = Fail	25%	5	10	10
		TOTALS:	100%	50,00%	87,50%	75,00%

[Nickel-hydrogen batteries for large-scale energy storage | PNAS](#)

[NiMH Battery, Nickel Metal Hydride Battery, NiMH Pack | Grepow](#)

[Comprehensive Understanding of Lithium-ion Battery Life Cycle \(cmbatteries.com\)](#)

Battery Trade Study						
Criteria	Explanation	Grade	Weight	1500 MAH nickel-hydrogen battery from Nuoxing	Eaglepic her 72 Ah Space Cell	Eaglepic her 55 Ah Lithium-Ion Cell

Energy Density	Battery must be able to store energy compactly and in a weight efficient manner	10 = <500 wh/kg 5 = < 150 wh/kg 1 = < 50 wh/kg 0 = Fail	25%	9	8	10
Temperature	Battery pack must be able to provide power at low temperature ranges	10 = -50 to 50 C 5 = -25 to 50 C 1 = 0 to 50 C 0 = Fail	25%	10	5	8
Reliability	Are the batteries reliable and how long does it take for them to fail? What happens when they fail?	10 = over 10000 cycles 5 = over 1000 cycles 1 = over 100 cycles 0 = Fail	25%	7	10	10
Discharge Voltage	Discharge rate and voltage	10 = low voltage change 5 = medium voltage change 1 = high voltage change 0 = Fail	25%	3	8	6
		TOTALS:	100%	72,50%	77,50%	90,00%

[SLC-21060-003 0322 0.pdf \(eaglepicher.com\)](#)

[LP32770 0223.pdf \(eaglepicher.com\)](#)

[Supply 1500 Mah Rechargeable Nickel Hydrogen Battery Outgoing Battery Aa Camera Battery No.7 Battery - China Wholesale Lithium Battery \\$0.43 from DongGuan NuoXing Electronic Co.,Ltd | Globalsources.com](#)

1.5.4. Command and Data Handling (CDH) Subsystem

The command and data handling subsystem allows for communication between all other subsystems, similar to a brain. It will both take in input from other subsystems, such as battery health and information from instruments, and then command other subsystems, such as telling the rover to move, telling the rover to recharge, or instructing the instruments to drill. Within the command and data handling subsystem itself, the components must also be able to work in unison, including the ability to store data computed, compute data received, and communicate data stored or computed.

1.5.4.1. CDH Subsystem Requirements

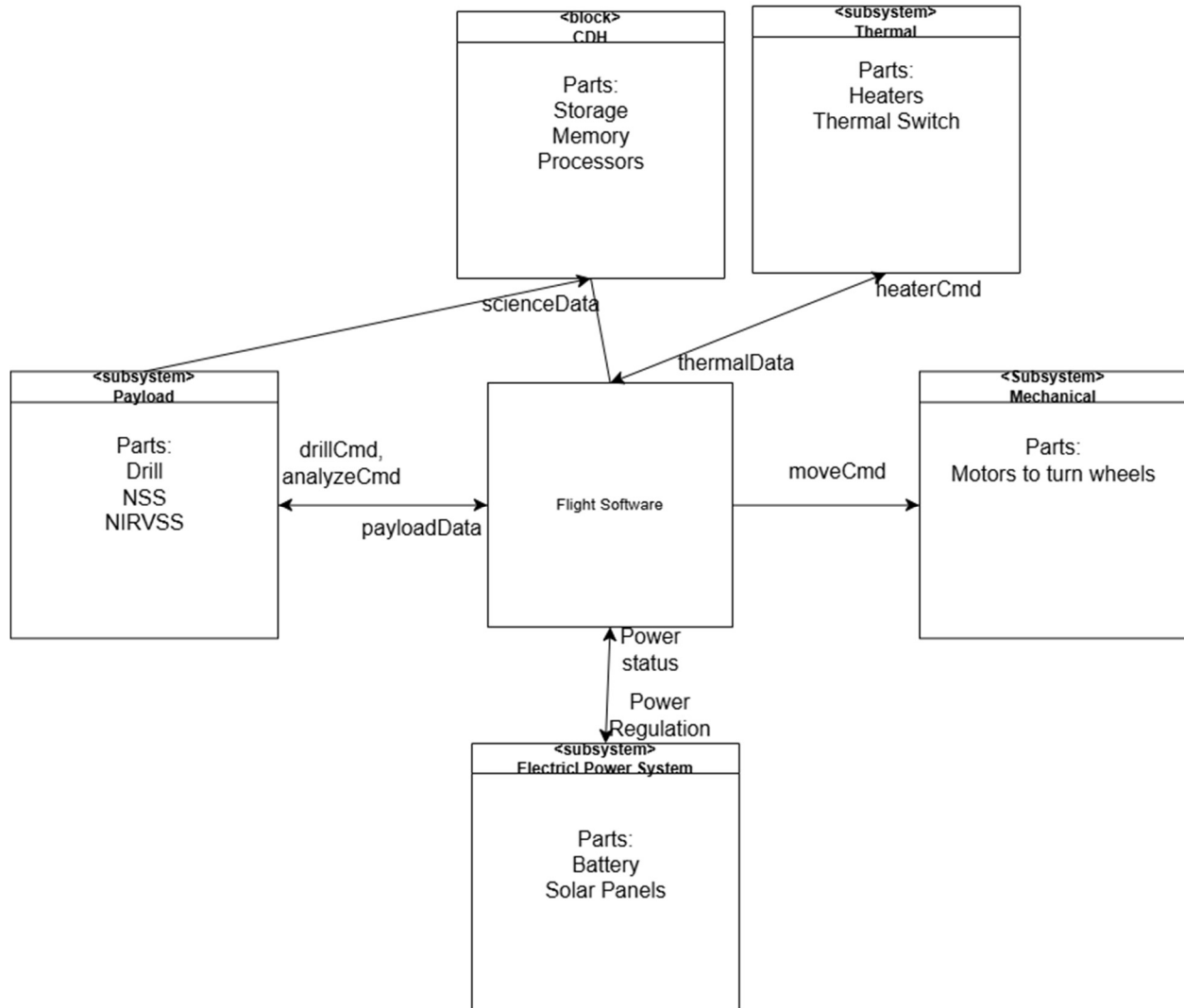
The command and data handling subsystem requirements are based on the ability for the various components needed from the command and data handling subsystem and the components and instruments from the other subsystems working together to send information in and out of the processor, storage, and the chosen communication method. The ability for the command and data handling subsystem creates its own requirements, as the components that create the ability to communicate need the ability to do so.

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
CDH-01	The CDH subsystem shall be able to withstand the lunar pole temperatures range of 90.15 K to 379.15 K	Temperature range is needed for operations to perform	TCS-1.2	N/A	Test	Thermal Control System	Not Met
CDH-02	The CDH subsystem shall be able to send communications to the primary mission orbiter.	Data must be available to NASA for observations	SYS-06	N/A	Test	Command & Data Handling	Met
CDH-03	The CDH subsystem shall be able to receive communications from a primary mission orbiter.	Rover must be able to be commanded by NASA when needed	SYS-06	CDH-03.1	Test	Command & Data Handling	Met
CDH-04	The CDH subsystem shall be able to store data and communications received	Allows for computation to happen	SYS-09	CDH-04.1	Test	Command & Data Handling	Met

CDH -05	The CDH subsystem shall be able to compute data onboard	Allows for data to be properly sent and reliable actions	SYS-09	N/A	Test	Command & Data Handling	Met
CDH -06	The CDH subsystem shall be able to use software to manage the various operations of the rover	Allows for all systems to work as intended	SYS-01	N/A	Test	Command & Data Handling	Met
CDH -09	The subsystems shall have a mass of no greater than 85 kgs.	System needs to be light enough to guarantee harmony with the other systems.	SYS-02	N/A	Test	System	Not Met
CDH -10	The CDH subsystem shall use no greater than 275 watts.	System needs to split power among the systems	PS-07	N/A	Test	Payload, Command & Data Handling	Not Met
CDH - 02.1	The CDH subsystem shall be able to send and receive communications at a distance of 100km	Orbiter will be within this radius.	CDH-02	N/A	Test	Command & Data Handling	Met
CDH - 04.1	The CDH subsystem shall be equipped with a storage device capable of performance that will not delay the rover	Storage must be large enough for all of the information.	CDH-04	N/A	Test	Command & Data Handling	Met
CDH - 06.1	The CDH subsystem shall be able to act autonomously when communication is not available	Allows for missions to PSR	0.1	N/A	Test	Command & Data Handling	Met
CDH - 06.2	The CDH subsystem shall be able to send and receive communications to and from the other subsystems	CDH is responsible for making sure the other subsystems are functioning	CDH-06	N/A	Test	Command & Data Handling	Met

		properly					
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1.5.4.2. CDH Subsystem Overview



The Command and Data Handling subsystem is the brains of the spacecraft that is being outlined in this report. This subsystem has several specifications that must be fulfilled by the commercial off the shelf (COTS) parts that are being selected. These specifications are found in section 1.5.5.1 of this report and include; survivability, communication with the rover from Earth, data storage, internal craft communication, weight specifications, and power draw. The pieces of equipment that will uphold these requirements are the central processing unit (CPU), communication antenna,

communication receiver, data bus, storage, and onboard software. The final selection of these components must meet the list of requirements found in section 1.5.5.1 but also must be able to operate on the lunar south pole.

Operation of the rover in this environment comes with its own set of difficulties for the CDH subsystem that should be addressed in regards to the requirements to ascertain solutions for these problems and to develop systems to mitigate or negate them in order for the system to perform at maximum capacity. These difficulties include environmental concerns such as radiation management for all components, and the activation of the CDH system in preparation for lunar operation.

First, the most detrimental environmental threat to the CDH system on the moon is its environmental radiation. On the lunar south pole the CDH system can expect to receive about 0.00139 sieverts or 0.139 rads of ionizing radiation per day. (AFP, 2020), This is important regarding the accuracy and function of the CDH system as ionizing radiation can flip computer bits from 1 to 0 when a transistor is open which can corrupt memory or introduce error into a calculation. This is called a single event upset, or (SEU) and occurs when an ionizing particle strikes a transistor in an “empty” state leaving behind a trail of ions that fills the transistor with electrons causing a bit to “flip” from a zero to a one. The opportunity or window for this event to occur depends on how often each transistor is in an open state, this is commonly represented as how often a transistor can be flipped within a second. This is dependent on the system's clock speed as the greater the number of cycles of the system each second matches the number of opportunities for a bit to be flipped each second. Therefore by lowering the system's clock speed the probability of data and calculation corruption will also be lowered. This will result in a slower but more accurate and reliable processor and will be combined with a depleted boron substrate on the CPU to further decrease the effects of ionizing radiation on the CDH system in the lunar rover.

For the remainder of the systems in the CDH subsystem, radiation ratings and specifications will suffice for the requirements of the craft. Outside of the adjustments to increase the performance of the craft, the components that make up the CDH subsystem must be able to uphold the requirements in section 1.5.5.1, therefore the selection of these components necessitates special attention. The primary component that all of the CDH system's components will connect in some way to will be the central processing unit (CPU).

The component we chose for this important part of the craft will be the RAD5545 CPU made by BAE systems. This CPU was selected because it satisfies multiple roles in the CDH subsystem, such as the motherboard, CPU, and graphics card, decreasing the complexity of the assembly and physical failure points of the system. The RAD5545 is a single-board computer (SBC) with a combined system on a chip, effectively combining

both the motherboard, processor and graphics card together which decreases the number of parts in the system and also simplifies the power supply for these components as it only needs one 5V connection to power the board rather than 3 separate power connection points for each of the subcomponents. On top of this the power consumption for this board is below 30 watts at peak performance, outmatching other SBC's for a low power draw.

Besides the physical simplifications that make this board advantageous it also outperforms in several other key areas. The first of which is its radiation hardness, while the RAD5545 may not be as radiation tolerant as other SBC's on the market, it far exceeds the radiation requirements for the environment on the moon, with a specified radiation tolerance of 100,000 rads which exceeds the 0.139 rad's of radiation received on the surface of the moon each day and the total number of rad's received for 40 days. which falls significantly below the 100,000 rad limit received for the mission timeline found in section 1.7.3.

There are several other characteristics that make this SBC advantageous compared to other options. This CPU operates at higher computation/clock speeds around 250 MHz compared to other SBC's which offers the advantage of faster processing times and quicker response times when processing data. In some ways this is not advantageous as higher clock speeds can increase the chances or opportunity for a bit to flip from a 0 to a 1 in a SEU as previously discussed, however depending on the error rate observed the CPU can be clocked down to decrease this error or kept at a higher rate to process more information faster offering more flexibility for the team depending on the amount of radiation hardening and processing power needed.

Outside of the CPU, an important component for internal data transfer and communication is the data bus. This component which makes up the wiring between the subcomponents of this system is necessary in order for the components to communicate with each other and operate effectively. For this purpose the CDH team selected Space Wire as our data bus. There are several advantages to using space wire. Primarily the radiation hardness that space wire provides is significantly greater than other options and supports a much better fault tolerant network than other data bus options like MIL-STD-1553 wire or regular ethernet cable. This is advantageous to our use case because the lower error rates that space wire provides because of its greater radiation hardness allows the system to function with less error as a whole, thus increasing the clarity of data transfer and decreasing operating errors. Outside of this, the total power consumption of the space wire system is significantly at around 50mW per driver/receiver pair compared to ethernet—which can use up to 30W for a driver/receiver pair. This makes space wire advantageous to our use case as it consumes less energy and therefore can support the same operations in a role where power consumption should be kept to a minimum.

There are several other components that are necessary for the rover to uphold its requirements outlined in the SU 2024 mission task and in the CDH requirements section 1.5.5.1 of this document. Two important components that are necessary for communication to and from the rover are the communications antenna and the communications receiver. The communication antenna that was selected by the CDH team was the BAE Systems Ka-band Multi Orbit LEO Optimized Transmitter in a half size array of 1 transmit antenna. This antenna was selected primarily because it has high rates of data transfer over the Ka band and uses the same radio band as the LRO. The power consumption and weight for this transmission antenna are approximately 55 watts for transmission or reception at peak operation and 2.2 kg. This fits the needs of this mission well as it has a low weight and power draw compared to other options. On top of this the transmitter is electronically steerable.

The second component that was selected for the communications array was the receiver, which the CDH subteam selected as the BAE Systems Ka-band Multi Orbit LEO Optimized Receiver in a $\frac{1}{3}$ size array of 2 receiving subarrays. These communication arrays were selected in the $\frac{1}{2}$ and $\frac{1}{3}$ sizes because the COTS components are size-optimized for LEO, coming in a large array for communication at distances of up to 800 km, thus exceeding the requirements for communication with the LRO which resides at 50 km above the lunar surface. Additionally, this array size fulfills both the transmitter and receiver requirements for the rover in section 1.5.5.1.

The last hardware piece of the CDH subsystem is the memory. For this system, the memory architecture that the CDH team chose was flash storage because of its low power requirements, superior memory stability, and vibration resistance in space environments. This makes flash memory a great option for the CDH subsystem and for this rover. After deciding the memory architecture the CDH team decided that the rover will have 500GB of data storage for images, sample and test information, and the operating system. This is because the craft will need to store information between transmission intervals with the LRO. This amount of storage will allow the rover to continue functioning when it is not in contact with the LRO and store information that is needed onboard at all times. On top of this flash storage is not heavy, space consuming, or energy intensive therefore the memory size of the storage module will impact the craft very slightly. This means that the memory size can be much larger than what is actually needed and not drastically impact the weight or volume requirements for the system.

The Operating System will be Linux, a well-established OS in space applications, such as the SpaceX Satellite network (Tung). On top of this, it is a commonly utilized OS in a multitude of non-space operations. Thus, there is a widely-supported support network for both the linux platform and for related hardware components. Outside of this, Linux is very streamlined, making it much more power-efficient than similar Operating

Systems—while also maintaining its versatility—as opposed to more dedicated space operating systems. For instance, Real Time Executive for Military Systems (RTEMS) is a known as an operating system designed for spaceflight applications, but is more strictly dedicated to rocket control. Because of these reasons the CDH team chose Linux as the operating system for the rover.

To conclude the overview of the CDH subsystem we will discuss the TRL of all subcomponents and of the whole system in conjunction with the weight, volume and power requirements and specifications of the whole system. Starting off, the SBC has a TRL of 6 as it has been designed for use in a space environment and tested on earth in a space like environment. The type of flash memory benign used has been used in space extensively over the years and has therefore been flight proven in a space environment, this earns it a TRL of 7. Moving on, the antenna, transmitter and data bus have all been built for and tested in space environments, this earns each of them a TRL of 6. Overall this subsystem when all of its components have been combined together will have a TRL of 6 as all of the components have been tested or proven in a space like environment but the whole system has not been tested in a similar manner, therefore to improve the systems TRL, testing must be done in a space like environment.

Component	Weight (kg)	Volume (L*W*H) (cm)	Power Requirements (watts)	TRL
Single Board Computer	0.24 kg	18.5 cm x 18 cm x 4.5 cm	30W max	6
Transmitter	7kg T+R	(10 cm x 20 x 5cm) T+R	55W upload	6
Receiver		(10 cm x 20 x 5 cm) T+R	65W download	6
Data Bus	0.03 kg/m	N/A		7
Storage	0.03 kg	5 x 2.5 x 1 cm ³	Write: 0.0067 W Read: 0.0027 W Off: 0 W	6
Software	N/A	N/A	Low	
Total:	7.57 kg	3507.8 cm ³	150.0094 W	

(12)

1.5.4.3. CDH Subsystem Trade Studies

On-Board Computer						
Criteria	Explanation\	Grade	Weight	RAD750® radiation-hardened PowerPC microprocessor	GR740 Quad-Core LEON4 FT SPARC V8 Processor	RAD5545
Durability	SYS-03	10 = More than adequate temperature range 7 = Adequate temperature range 3 = Not adequate temperature range 0 = Very poor temperature range	25%	7	7	7
Speed	The CPU must be able to operate at high speeds in order to guarantee maximum efficiency.	10 = > 250 mHz 7 = 200-250 mHz 3 = 100-200 mHz 0 = below 100 mHz	25%	7	7	10
Radiation Hardness	The CPU must be able to survive the high-radiation environment of the lunar poles.	10 = TID > 500kRad 7 = TID >= 100 krad 3 = TID below 100krad 0 = no radiation hardening	25%	10	7	7
Power Consumption	Power consumption should ideally be as low as possible to save room for other	10 = below 30 watts 7 = 30-40 watts	25%	7	10	10

	components.	3 = 40-50 watts 0 = 50 watts				
		TOTALS:	100%	77.50%	77.50%	85.00%

Data Bus						
Criteria	Explanation	Grade	Weight	Space Wire	MIL-STD-1553	Ether net
Speed	The bus should be able to communicate using high-speed data transfer.	10 = large range, 7 = medium range, 3 = low range, 0 = none	25%	7	3	10
Radiation Hardness	Bus must be able to withstand lunar pole radiation.	10 = high, 7 = medium 3 = low 0 = none	25%	10	10	3
Power Consumption	The bus should make efficient use of the available power.	10 = very low, 7 = low 3 = medium 1 = high	25%	10	5	7
Error Rates	The bus must make as few errors as possible.	10 = very low, 5 = low 1 = medium 0 = high	25%	10	10	10
		TOTALS:	100%	92.50%	70.00%	75.00%

[D3285_KA+Band+SATCOM+Antenna+REBRAND.pdf](#)

Antenna						
Criteria	Explanation	Grade	Weight	X-Band High-Gain Antenna (NASA Perseverance)	Ultra-High Frequency Antenna (NASA)	X-band Low-Gain Antenna (NASA Perseverance)

				e, "Perseverance Rover Components.")	Perseverance, "Perseverance Rover Components.")	ce, "Perseverance Rover Components.")
Distance	CDH-02.1	10 = Communication far beyond 100km 8 = Communication to 100km reliable 4 = Communication to 100km unreliable 0 = Unlikely to communicate	35%	8	10	10
Reliability	CDH-02	10 = Proven Reliability 5 = Unproven Reliability 0 = Unreliability	25%	10	10	10
Transmission Rates	CDH-02.1	10 = Most transmission 7 = Good transmission 0 = No transmission	30%	10	7	0
Ease to Store Data	CDH-04	10 = No issues storing 5 = On device issues may occur 0 = Fail	10%	10	10	10
		TOTALS:	100%	93.00%	91.00%	70.00%

Storage						
Criteria	Explanation	Grade	Weight	Flash Storage (NASA Perseverance, "Perseverance Rover Components")	SSD (J.N. Maki et al., "The Mars 2020 Engineering Cameras and Microphone on the Perseverance Rover: A Next-Generation Imaging System for	Read-Only Memory (NASA Perseverance, "Perseverance Rover Components")

				ts.")	Mars Exploration.")	.)
Reliability	CDH-04	10 = Proven Reliability 5 = Unproven Reliability 0 = Unreliable	25%	10	8	10
Size	CDH-04.1	10 = Ample Storage 8 = Storage to meet requirements 0 = Not enough storage	25%	8	10	0
Speed	CDH-04	10 = Quickest Operations 6 = Proven speed, though not quickest 0 = Not enough storage	25%	10	10	10
Ease of Communication	CDH-04.1	10 = Easy to communicate with 5 = Requires additional bus to CPU 0 = Fail	25%	10	5	10
		TOTALS:	100%	95.00%	82.50%	75.00%

Software Trade Study						
Criteria	Explanation	Grade	Weight	VxWorks operating system	Linux Operating system	RTEMS - Real- Time Executive for Military Systems

Power Usage	How much power does it use? Is it efficient?	10 = high, 5 = medium 1 = low 0 = Fail	16%	8	5	5
Memory Management	Can we directly control the management of memory to improve performance?	10 = high, 5 = medium 1 = low 0 = Fail	20%	8	9	5
Radiation Resistance	Is the receiver radiation and single upset event tolerant?	10 = high, 5 = medium 1 = low 0 = Fail	16%	1	1	1
Operating Architecture	Does it run on a 256-bit system? 64 bit? How accurately can it represent information?	10 = high, 5 = medium 1 = low 0 = Fail	16%	8	8	5
Support	How large is the support network? Is it open source?	10 = high, 5 = medium 1 = low 0 = Fail	16%	6	10	6
Applicability	Is the operating system specifically designed for the application?	10 = high, 5 = medium 1 = low 0 = Fail	16%	9	7	7
		TOTALS:	100%	67,20%	67,60%	48,40%

Transponder (Receiver) Trade Study						
Criteria	Explanation	Grade	Weight	Small Deep-Space Transponder (SDST), Built by General Dynamics	SWIFT KTX, Built by Tethers Unlimited	SWIFT KTRX, Built by Tethers Unlimited

Power Usage	How much power does it use?	10 = high, 5 = medium 1 = low 0 = Fail	16%	5	10	1
Radio Frequency	Does it use the S or Ka RF bands? (must)	10 = high, 5 = medium 1 = low 0 = Fail	20%	10	10	10
Temperature Tolerance	What is the operational temperature range of the receiver?	10 = high, 5 = medium 1 = low 0 = Fail	16%	10	5	5
SEU / Radiation Tolerance	Is the receiver radiation and single upset event tolerant?	10 = high, 5 = medium 1 = low 0 = Fail	16%	10	5	5
Size and Weight	How large and heavy is the instrument	10 = high, 5 = medium 1 = low 0 = Fail	16%	5	10	10
Moving Parts	Are there any moving parts? (This may impact reliability).	10 = high, 5 = medium 1 = low 0 = Fail	16%	10	10	10
		TOTALS:	100%	84,00%	84,00%	69,60%

1.5.5. Thermal Management Subsystem

The thermal management subsystem's goal is to allow for all rover operations to occur on the lunar pole. Due to the objective of exploring a PSR of the lunar surface, the rover must have the ability to handle cold temperatures (as compared to the temperatures most Earth based rovers would experience) while maintaining normal operations, or creating conditions in which the rover knows how to operate, given the thermal conditions.

1.5.5.1. Thermal Management Subsystem Requirements

The thermal management subsystem requirements aim to create an operating temperature window in which the rover will operate, allowing for an understanding of how all subsystems will behave in the thermal environment of the lunar pole and PSR, meaning that decisions regarding the rover can be made accordingly. The ability to

regulate the rover's internal temperature despite changes externally is at the core of what the thermal management subsystem defines:

Thermal Control System Reqs (TCS)	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
TCS-1	The system shall maintain an internal temperature of 20 ± 15 degrees Celcius at all times until the end of the mission.	Ensure systems are able to properly function.	MR-1	TCS-1.1, TCS-1.2	Test	Thermal Control System	Not Met
TCS-1.1	The system shall maintain the defined internal temperature range when at max and minimum power.	Ensures system functions properly at power extremes.	TCS-1	N/A	Test	Thermal Control System	Not Met
TCS-1.2	The system shall maintain the defined internal temperature range in the coldest and hottest lunar surface temperatures (90.15 K to 379.15 K)	Ensures system functions properly at temperature extremes.	TCS-1	N/A	Test	Thermal Control System	Not Met
TCS-2	The system shall not cost greater than the amount of money allocated to the thermal system.	Ensures there is enough money to design, build, and test the system.	TCS-1	N/A	Analysis	Thermal Control System	Not Met
TCS-3	The system shall have a thermal model that shows net heat transfer equal to zero.	Ensures the system accounts for all forms of heat transfer throughout the design process.	TCS-1	TCS-3.1 TCS-3.2	Inspection	Thermal Control System	Not Met
TCS-3.1	The thermal model shall include the main forms of heat transfer in space, including conduction and radiation.	Ensures the system accounts for all forms of heat transfer throughout the design process.	TCS-3	N/A	Inspection	Thermal Control System	Not Met
TCS-3.2	The thermal diagram shall model extreme hot and cold cases.	The hot extreme will be at the hottest surface temperature and full power of components. The cold extreme will be at the coldest surface temperature with the system at low power. This ensures the thermal system accounts for the worst	TCS-3	N/A	Inspection	Thermal Control System	Not Met

		extremes.					
TCS-4	Exposed exterior components such as solar panels shall survive extreme lunar surface temperatures from 90.15 K to 379.15 K throughout the duration of the mission.	Ensures components not contained within controlled temperature in the instrument can survive extreme temperatures.	TCS-1	N/A	Analysis	Thermal Control System	Not Met
TCS-4	The internal thermal system shall not exceed internal structural dimensions.	Ensures the thermal system fits within volume constraints.	TCS-1	N/A	Analysis	Thermal Control System	Not Met

1.5.5.2. Thermal Management Subsystem Overview

The thermal subsystem regulates the internal temperature of the rover to protect internal electronics from extreme lunar surface temperatures. The main components to mitigate extreme temperature fluctuation in the rover are a heater, a radiator, and a thermal switch. The heater will keep electronics from getting too cold in PSR and the radiator will keep the internals cool when in sunlight. The thermal switch automatically regulates the thermal system between temperature extremes by switching on the heater if the temperature is too low. The thermal switch is a passive device, allowing the system to adjust to the surroundings automatically. In addition to these components, a sprayable thermal control coating can be applied to the surface as a passive temperature regulator.

The heater chosen for the rover is the Omega KHLVA, PLM-Series. It comes in a variety of sizes ranging from 1-5 inches in Length and Width. All heaters are 0.01 in thick and can be mounted on curved surfaces due to their flexibility. These heaters can operate in -57 to 232°C without adhesive and -40 to 149°C with adhesive.

The Diaphragm Thin Plate is the recommended device that allows the rover to switch a heat conduction path of strong and weak thermal coupling. Sierra Nevada develops Diaphragm Thin Plates that have a max conduction of 607 W/m²/°C and the minimum conduction of 7.1 W/m²/°C. The size is also relatively small, which becomes relevant to the design of the spacecraft itself. Compared to the alternative options of the thermal switches which are— a Passive Thermal Control Heat Switch and the HP Thin Plate, the Passive Thermal Control Heat Switch has a low max and minimum conductivity, however, the HP Thin Plate comes extremely close to a match in our trade studies. Both the Diaphragm Thin Plate and the HP Thin Plate have similar conductivity and volume.

Radiators within lunar exploration provide heat rejection for rovers or spacecraft for the experience through the lunar harsh atmosphere and environment. Active Thermal Architecture (Utah State University) allows flexibility of the design with the factors of versatility with the high performance deployable MFPL radiator. Unlike the Q-Rad and the Triangular Fins radiators, the Active Thermal Architecture's max TRL is higher with a wide range of operating temperature range and has is relevantly small.

A sprayable thermal coating is an additional option to aid the surface of the rover in radiating heat. While this paint can be expensive, its ability to passively radiate heat from the spacecraft helps protect the internal electronics from hot surface temperatures. Based on the trade study, the Socomore A276 Polyurethane white paint is the best option, mainly due to its lower cost. It was also rated as an easy to apply paint, which will shorten the manufacturing time.

Table 0. Thermal System Subcomponents

Component	TRL	Mass (kg)	Volume	Max Power Draw (V)
Omega KHLVA, PLM-Series (Heater)	6	0.1 lb/in ³ (2.72 g/cm ³)	1 x 12 x 0.01 in ³ Note: Comes in varying sizes	115
Active Thermal Architecture (Utah State University)	7	N/A	TBD	TBD
Sierra Nevada Thin Plate Heat Switch	7	N/A	1 x 1 x 0.32 in ³ (25.4 x 25.4 x 8.13 mm ³)	TBD
Sprayable Thermal Control Coating	6	N/A	1 gallon (The entire bucket will likely not be used)	N/A
Q-Rad Deployable Radiator	5	N/A	10 cm x 20 cm x 3 cm	TBD

1.5.5.3. Thermal Management Subsystem Trade Studies

Table 0. Heater Trade Study

Heaters						
Criteria	Explanation	Grade	Weight	Omega KHLVA, PLM- Series	Tempco SHK Series	Minco Polyimide Thermofoil HK Series
Watt Density	A Watt density provides more heating power per unit area.	10 = high, 5 = medium 1 = low 0 = Fail	15%	2	8	6
Temperature Limits	A greater temperature range will ensure the survivability of the component.	10 = high, 5 = medium 1 = low 0 = Fail	50%	6	3	7
Cost	A lower cost will allow for more funds to be allocated to other subcomponents and subsystems.	10 = high, 5 = medium 1 = low 0 = Fail	10%	9	2	4
Size	A smaller size will allow for more room for other components in the internal structure. This also allows greater versatility in positioning the thermal system.	10 = high, 5 = medium 1 = low 0 = Fail	25%	8	6	4
		TOTALS:	100%	62.00%	44.00%	58.00%

Table 0. Radiator Trade Study

Criteria	Explanation	Grade	Weight	Redwire Space : Q- Rad	Active Thermal Architecture (Utah State	BYU: Triangular Fins
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					Unive rsity)	
Power (W)	The less Watts a radiator uses is more reliable and efficient for the project as there are many components going into the design -- plus the spacecraft is small.	10 = high, 5 = medium 1 = low 0 = Fail	30%	7	8	7
Area (M^2)	The area of the radiator may affect the design and assembly of the spacecraft. A lower area will allow space for other sub-components.	10 = high, 5 = medium 1 = low 0 = Fail	15%	8	9	7
Temperature	A wide range of operating temperatures will allow flexibility on the radiator in extreme lunar weather and environment.	10 = high, 5 = medium 1 = low 0 = Fail	30%	8	7	7
Cost	Reduced cost allows funds to become more flexible for the many components that the spacecraft requires.	10 = high, 5 = medium 1 = low 0 = Fail	25%	N/A	N/A	N/A
		TOTALS:	100%	74.50%	76.00%	70.00%

Table 0. Thermal Switch Trade Study

Thermal Switch						
Criteria	Explanation	Grade	Weight	Passive Thermal Control Heat Switch	HP Thin Plate	Diaphragm Thin Plate
Conductance	Balances efficient heat transfer during extreme temperature (preventing overheating and insulates during temperature drops) variations while also providing durability and reliability in the harsh lunar environment.	10 = high, 5 = medium 1 = low 0 = Fail	40%	5	8	9
Max Q	Maximum dynamic pressure that the rover experiences during launch to the Moon and during any maneuvering that might induce significant aerodynamic forces.	10 = high, 5 = medium 1 = low 0 = Fail	40%	6	8	8
Volume	Low volume allows flexibility of the design process and structure for the other sub-components.	10 = high, 5 = medium 1 = low 0 = Fail	10%	8	9	8
Mass	Less density the better to allow flexibility on structure for other sub-components.	10 = high, 5 = medium 1 = low 0 = Fail	10%	8	7	7
		TOTALS:	100%	60.00%	80.00%	83.00%

Table 0. Sprayable Thermal Control Coating Trade Study

Sprayable Thermal Control Coating

Criteria	Explanation	Grade	Weight	Socomore A276 Polyurethane	Huntington Ingalls Industries Z93-C55 Silicate	AZ Technology AZ2000 Silicate
Cost	This subcomponent is a secondary measure in addition to a dedicated radiator. A low cost paint should be applied to ensure there is enough money for other subcomponents.	10 = high, 5 = medium 1 = low 0 = Fail	15%	8	4	4
Difficulty to Apply	The ease of application will speed up the manufacturing process. This helps keep the entire project on schedule.	10 = high, 5 = medium 1 = low 0 = Fail	15%	10	3	3
Durability	Must be durable in the harsh space environment. Must be resistant to UV. Any silicate based paint requires temperature control, which could negatively affect the durability.	10 = high, 5 = medium 1 = low 0 = Fail	35%	5	8	6
Color	The paint must be a white color to make the surface area of the rover act as a radiator.	10 = high, 5 = medium 1 = low 0 = Fail	30%	10	10	7
		TOTALS:	95%	74.50%	68.50%	52.50%

1.5.6 Payload Subsystem

The payload subsystem shall be able to collect 100g of ice samples as well as be able to determine the water ice abundance at one of the 13 Artemis III landing sites. Due to the environment at these landing sights, we will need to be able to have the payload withstand the harsh conditions.

To accomplish these, the spectrometer box holds one short-wave (1200-2400 nm) and one long-wave (2300-4000 nm) spectrometer. The short-wave spectrometer has a spectral resolution of <20 nm, and the long-wave <50 nm. Both spectrometers are fiber-optic fed.

In addition, the bracket assembly consists of four parts that bring the total mass of the instrument to 3.57kg.

1.5.6.1 Payload Subsystem Requirements

Req #	Requirement	Rationale	Parent Req	Child Req	Verification method	Relevant Subsystem	Req met?
PAY-01	The subsystem shall survive the extreme temperatures of the lunar pole ranging from 90.15 K to 379.15 K	In order to complete science tasks the instruments must be able to withstand the environment	0.1, SYS-03, SYS-07	TBD	Test	Thermal	Met
PAY-02	The subsystem shall be mounted on the spacecraft where the line of sight is clear of any obstructions	To collect data efficiently the instruments will need an open view of the lunar surface	0.2	TBD	Inspection	Mechanical	Met
PAY-03	The subsystem shall consist of two instruments that will directly take part in scientific observations	The two instruments will perform the necessary tests to obtain data	SYS-05	TBD	Verification	Mechanical	Met
PAY-04	The subsystem shall have sufficient power to operate	In order to run the instruments power is required	SYS-07	TBD	Test	Electrical	Met
PAY-05	The subsystem shall detect volatiles using a short and long wave ranging from 1200-2400 nm and 2300-4000 nm respectively	To successfully complete tasks certain wavelength parameters must be to par	0.2, SYS-05, SYS-09	TBD	Test	Electrical, CDH, Computer hardware	Met

PAY-06	The subsystem shall be able to collect 100 g up to 1 m depth of lunar regolith	To interpret the abundance of volatiles under the surface, beneath surface samples will contribute	0.2, SYS-05, SYS-09	TBD	Test	Mechanical	Met
PAY-07	The subsystem shall store all collected data for relay to the LRO	After samples and data has been analyzed, it must be relayed back to earth for analysis	SYS-05, SYS-06	TBD	Test	Electrical, Computer hardware, CDH	Met

1.5.6.2 Payload Subsystem Overview

The payload subsystem will focus on detecting volatiles within the lunar regolith. It will do so using two scientific instruments. The first of which is to determine where volatiles, as determined by detecting hydrogen, are present within the lunar regolith. The first instrument chosen to determine where hydrogen is present is an NSS as this instrument was determined to be able to continuously run, reliably detect hydrogen, and have a high survivability. This first step will determine areas of interest for further inspection.

The next step involves the collection mechanism to obtain the samples for further study. The sample collection method of choice is a TRIDENT drill as it was discovered during a trade study that the instrument is reliable, able to collect scientifically interesting samples, as well as capable of reaching the requisite depth required to demonstrate volatile compounds within regolith samples.

After obtaining the sample of interest, the second instrument within the payload consists of a NRVSS as this was determined to provide the required particle measurements needed to observe volatiles with enough detail, provide reliable service in the lunar environment, as well as be able to fit on the rover.

Through the use of spectrometers from instruments 1 and 2, the rover will be able to detect the chemical compositions of sampled regolith and determine if any volatiles are present to a fine degree of accuracy.

1.5.6.3 Payload Subsystem Trade Studies

Instrument 1 (finding hydrogen)					
Criteria	Explanation	Grade	Weight	NSS	DAN
Continual Operation	The first instrument must be able to create locations for the second instrument to get more in depth readings.	10 = meant to continually run to map hydrogen 9 = can continually run 3 = slow run 0 = must stop	60%	10	9
Reliability in detecting hydrogen	The first instrument needs to be able to detect hydrogen and report it back with a high degree of confidence	10 = low margin of error 0 = unreliable	20%	10	10
Survivability	This instrument has been shown to be reliable at its function in space	10 = reliable in space mission 0 = unreliable	20%	10	10
		TOTALS:	100%	100.00%	94.00%
				NASA, "VIPER Rover and Instruments."	NASA, "Curiosity Science Instruments."

Instrument 2 (Analyze Hydrogen)						
Criteria	Explanation	Grade	Weight	NIRV SS	SAM	APXS
Depth of Analysis of Sample	The instruments being compared here will need to run in depth analysis of the samples collected, with this comparison allowing for conclusions to be drawn	10 = full analysis of composition of sample 7 = good analysis of composition of sample 2 = inadequate analysis of sample 0 = useless	40%	9	10	6

		analysis				
Survivability	This instrument has been shown to be reliable at its function in space	10 = reliable 0 = unreliable	15%	10	10	10
Ease of Analysis	Based on the instrument's ability to fit onto the rover	10 = proven use of instrument in analysis 0 = not proven for mission	10%	10	10	10
Particle Analysis Relevance to Mission	This category essentially asks the question: Does this instrument do too much? Extra function creates unnecessary complexity for the mission	10 = fulfills mission requirements of analysis for H2O/particle composition 6 = does too much or too little 0 = unrelated instrument	35%	10	7	6
		TOTALS:	100%	96.00%	89.50%	70.00%
				NASA, "VIPER Rover and Instruments."	NASA, "Curiosity Science Instruments."	NASA, "Alpha Proton X-Ray Spectrometer (APXS)."

Drill for Samples					
Criteria	Explanation	Grade	Weight	TRIDE NT	Percussive Drill
Reliability	The drill needs to survive for the duration of the mission without breaking. Judges instruments based on past performances.	10 = reliable in space mission 4 = shown as occasionally reliable	60%	10	6

		0 = unreliable			
Ease of collection	Based on the instrument's ability to integrate the drill into the rover	10 = integration will not require much specific accommodations 0 = difficult to put on rover	10%	10	10
Depth	The rover must be able to get samples from various depths in order to get various samples	10 = 1 meter depth 3 = shallow depth 0 = no depth	30%	10	3
		TOTALS:	100%	100.00%	55.00%
				NASA, "VIPER Rover and Instruments."	Webster, "Mars Science Laboratory: Curiosity Stretches Its Arm."

1.5.7 Recovery and Redundancy

Mechanical Recovery and Redundancy

The mechanical subsystem designed will have multiple points to ensure the successful operation of the vehicle given unforeseen circumstances such as obstacles in the terrain, multiple wheel design where failures can be mitigated by the suspension and drivetrain of the spacecraft. The structure design will also have multiple layers for thermal and strength redundancy. The mechanical subsystem will include the two science instruments to scan the lunar surface offering recovery if one system happens to be inoperable.

Power Recovery and Redundancy

The Power Subsystem can include multiple redundant components to ensure its reliability and ability to recover if a fault in the system occurs. All of the components in the power subsystem are critical to the mission and should they fail before mission completion they would result in mission failure. As some of the components are more prone to failure like the solar charge controller, inverter and power supply, these will be focused on to improve the systems reliability as they contain many electrical components that can burn out or fail as opposed to the batteries and solar panels which are heavy which makes it harder to improve the reliability by adding components in parallel, and they both utilize observed chemical processes that are

unlikely to fail unless physics breaks down. Therefore to improve the reliability of this system there are several steps we can take. The first is adding components in parallel so if one fails another can take over. This is disadvantageous for heavy components because it ensures that one will not be “dead weight” so it will be reserved for lighter components like the inverter. Otherwise, for components that are heavier it will be more advantageous to select components that are overbuilt for the job that they are tasked with. Combining both of these strategies will improve the reliability, redundancy and self recovery of the craft.

Thermal Recovery and Redundancy

The thermal system can include multiple heaters for redundant heating. The heater chosen from the trade study is less than \$100, flexible, and small, making it an easy redundancy measure to implement. If one heater fails, the others should keep the internal components warm enough to function. To reject heat from the surface of the rover, both a radiator and thermal coating are used in the thermal system. The thermal control coating acts as a redundant backup in case it fails or is inefficient, though this is not likely.

Command and Data Handling Recovery and Redundancy

The main computers can utilize error checking and error correcting algorithms to ensure proper computation and data handling. Similarly, duplicate computational systems can be used to ensure redundancy in data handling and processing. Systems, such as those indicated in the trade studies, can also be chosen for radiation hardness which will mitigate risk further. Communication issues can be mitigated by utilizing antennas which have been rated for space as well as incorporating systems that will attempt to make contact should communications be temporarily suspended. The size, weight, and cost of the antenna and transponder makes it unrealistic to include redundant components, so those will be documented as Single Point Failures should any malfunction occur.

1.5.8 Interface Control

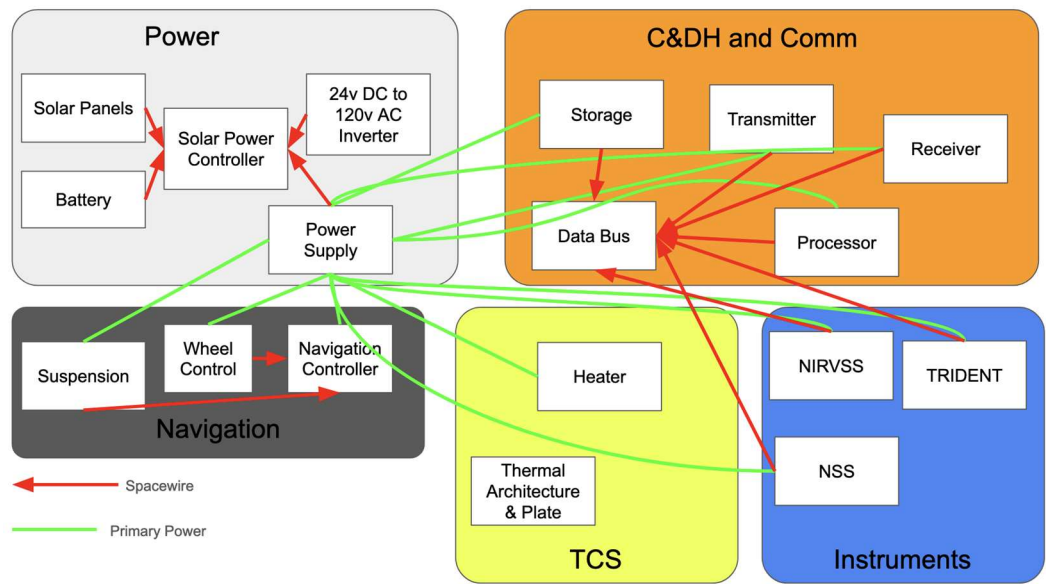
The interface between payloads is based around communicating only the necessary data from each subsystem. The data handling will communicate to all other subsystems, and all subsystems will communicate to the data handling subsystem. This creates the basis for how the communication will work, as most communications will happen this way. Furthermore, power is given to all other subsystems, though this should be obvious because all subsystems need power for at least one component or instrument. This structure creates a simple way of communicating, meaning that everything runs in and out of data handling, and power flows out of that subsystem.

N^2 Diagram

Payload	↕instrument info				
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☒ instrument commands	Data Handling	↘ instrument data, rover telemetry	↘ location to go to	↘ needs of subsystems temperature	↘ when to get more power
	☒ comms to rover	Comms			
	☒ telemetry		Navigation		
☒ when samples are collected, specific thermal	☒ current temperature			Thermal	
☒ Power	☒ Power	☒ Power	☒ Power	☒ Power	Power

Block Diagram



1.6. Risk Analysis

The thermal system is highly important for the transportation through the lunar’s atmosphere– the temperature, the lunar dust and the harsh environment overall. The risk can lead to the system to overheat causing damage to other subcomponents to the spacecraft itself.

SRR Risk Summary

ID	Summary	L	C	Trend	Approach	Risk Statement	Status
----	---------	---	---	-------	----------	----------------	--------

1	Mechanical	2	5	NEW	M	Wheels of rover fail to endure unexpected rocky terrain	Active
2	Mechanical	3	5	NEW	M	Solar arrays do not deploy	Active
3	Power	3	5	NEW	M	Solar cells are kept in a PSR for too long leading to a loss of power	Active
4	Power	2	2	NEW	M	Lunar dust particles settling on solar panels preventing efficient charging of the rover	Active
5	CDH	3	5	NEW	M	Radiation exposure causes damage to semiconductor devices leading to corrupted data or component, instrument, or vehicle loss or damage	Active
6	CDH	2	5	NEW	M	An event occurs which causes the rover to cease communication	Active
7	Thermal	1	4	↓	M	To prevent the system from overheating or freezing	Active
8	Thermal	2	5	↓	M	Radiators help regulate temperature through lunar environment	Active
9	Instrumentation	1	5	NEW	M	Thermal regulation system fails to keep the NIVRSS and NSS at appropriate operating temperatures	Active
10	Instrumentation	2	4	NEW	A	The LRO fails to fly over the rover to relay signals back to Earth	Active

L = Likelihood (1-5)		LxC Trend		Approach		Criticality
1 = not likely		↓ - Decreasing (improving)		A - accept		HIGH
5 = extremely likely		↑ - increasing (worsening)		M- mitigate		MED
C = Consequence (1-5)		→ - unchanged		W - watch		LOW
1 = low consequence		NEW - added this month		R - research		
5 = high consequence						

LIKELIHOOD	5					
	4					
	3					3, 5
	2		4		10	1
	1					9
CONSEQUENCES						
		1	2	3	4	5

(1) Given that the terrain of the Lunar surface near the south pole could be harsh near Haworth Crater, there is a possibility of the rover's wheels being damaged, resulting in unreliable equipment and lost mobility of the rover, thus ending the ability to collect data samples.

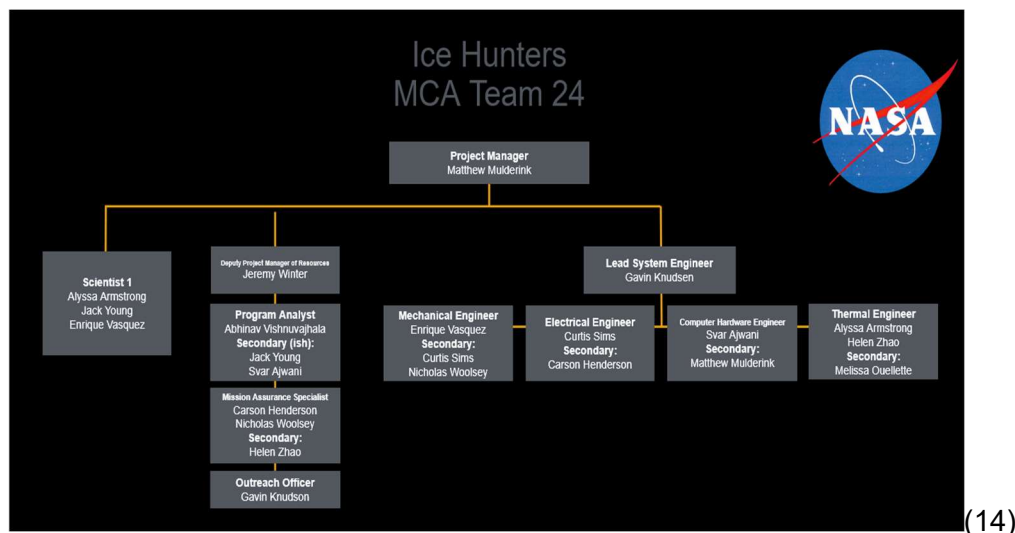
(3) Given that the rover is exploring areas within a PSR, there is a possibility that the solar cells go too long without a charge, resulting in a total loss of power, thus ending any ability to use the rover. (3) Given that the rover will be kicking up dust from driving and operating it's drills, there is a possibility that lunar dust will accumulate on the solar

panels, resulting in a less efficient power capture from the solar cells, thus decreasing the efficiency of the solar system.

(9) Given the operating temperature of the NIVRSS is 10° to 35° in the moon environment that ranges from -183° to 106° , there is a possibility that the NIVRSS will overheat or freeze over, resulting in damaged components, thus terminating the ability to analyze the composition of samples. (10) Given the LRO has a large flight path and unexpected re-directions can occur, there is a possibility that the LRO becomes off path, resulting in no communication between the rover and Earth, thus halting the rover's ability to move forward with drilling.

1.7. Programmatics

1.7.1. Team Organization



The team handles the workload, organization, and decision making in different ways. The workload is equally divided among the respective members of the team. The team is organized by people's specialities as well as prior experiences. The decision making is primarily left from person to person, however in the end it is the lead's responsibilities to make the final decision and be a potential tie-breaker.

Some of the challenges that the team overcame was having everybody be a contributor. Some people did the majority of the work while others did very little which was very unfair to those who did other people's work. For the future we will need to ensure that others are held accountable. For instance if someone does not complete what was asked of them, then we may have to turn to a higher authority to inform them of their actions.

Furthermore, team roles were all changed from the last deliverable, due to a mess up in assigning the roles and from a change in team member's availability to the team. There are now secondary roles, a reduction in the number of roles, and a new project manager.

1.7.2. Cost Estimate

The below cost analysis for the Ice Hunter mission outlines how the mission estimates it will allocate the total budget of \$225 million received in funding. All current estimates fall within acceptable ranges, based on past missions similar in scope.

Mission Class: Discovery (< 500 million)

[Link to budget template](#)

Justifications

- (1) Personnel: The justifications for the personnel distribution across phases were based on NASA's Project Management handbook, which outlined specifics for staffing requirements in the various project lifecycle phases.

Phase C has relatively fewer team members since it mainly consists of design work, but includes core members in all disciplines. Phase D has the most amount of team members due to the required intensive labor for building, integrating, and testing components. Phase E and F require less members than D since the labor is reduced, but more than C to ensure that all data is analyzed and operations run smoothly.

- (2) Travel: Assuming that team members are located in Johnson Space Center, where spacecraft missions are often conducted, a coach flight is, on average, than \$400 roundtrip to Cape Canaveral, where the launch is happening. For simplicity, I used an estimate of \$400 for all travel-related purposes.

According to the U.S General Services Administration's website, the September 2023 average rate for a one-night hotel booking is \$144, which was adjusted to a 2024 rate of \$148 (due to inflation).

For transportation, the team allocated an on-average \$50 per-day rental car so that up to 4 team members can travel in one car at a time.

The General Services Administration 2024 reports per-diem costs as \$163 for non-travel days and \$122.25 (75%) for travel days in a funded trip.

The travel specifics will be listed in the schedule portion of the SRR. Annual meetings, major milestone meetings, and team morale meetings are all considered in the budget.

- (3) NASA L'Space Project Management training recommends that 1% of the total mission budget is allocated towards outreach activities. The specific allocation of outreach materials and the percentage of resources set aside for each fiscal year was based on an average of multiple historical NASA missions.

Fiscal Year Allocation Split:

- FY1: 5%
- FY2: 10%
- FY3: 15%
- FY4: 20%
- FY5: 50%
- FY6: 5%

Outreach Category Split:

- Materials: \$3,000,000
- Venue Costs: \$2,000,000
- Travel Costs: \$1,250,000
- Services Costs: \$2,000,000
- Personnel Costs: \$2,000,000
- Total Margin: \$1,000,000

As this allocation exceeded 1% of the total budget due to inflation, the actual numbers in the cost template were modified to be slightly lower.

- (4) Direct Costs: The subsystem costs, manufacturing facility cost, and test facility cost were completely determined by NASA L'Space's MCCET, a parametric cost estimation tool.

Furthermore, the MCCET recommends that 4% of direct costs should be set aside for Phase B, 57% for Phase C, and 39% for Phase D. However, for Team Ice Hunters, Phase B is not included for budget estimates. So the MCCET recommendation was averaged with data from the VIPER mission and NASA handbook guidelines to provide this breakdown:

Phase Allocation

- C: 57%
- D: 39%
- E: 3%
- F: 1%

For each fiscal year, the split was determined by how much of the phase occurred in each year. For example, the mission starts in September of FY1, so it requires proportionally less funds than FY2. Also, FY3 includes both Phase C and D, so it requires more funds than FY4.

Fiscal Year Allocation

- FY1: 18.5%
- FY2: 28.5%
- FY3: 28%
- FY4: 21%
- FY5: 3%
- FY6: 1%

Subsystem Allocation (estimates provided from MCCET):

- Mechanical: \$7,556,887.54
 - *mass and maximum power provided in SRR*
 - Manufacturing costs: \$2,443,112.46
 - Test costs: \$3,000,000.00
- CDH: \$735,430.74
 - *mass provided in SRR*
 - Manufacturing costs: \$264,569.26
 - Test costs: \$300,000.00
- Power: \$21,421,718.99
 - *mass provided in SRR*
 - Manufacturing costs: \$7,178,281.01
 - Test costs: \$8,600,000.00
- Thermal: \$2,548,113.97
 - *mass estimated with similar industry-standard materials (4.1 kg)*
 - Manufacturing costs: \$751,886.03
 - Test costs: \$1,000,000.00
- Payload: \$38,602,641.18
 - *Used team data for mass, averaged multiple lunar missions for lifecycle estimate of 1 year, averaged Curiosity and VIPER missions for maximum power (287.5 W)*

- Manufacturing costs: \$13,297,358.80
- Test costs: \$15,600,000.00

For total manufacturing and test costs, a similar fiscal year breakdown to the subsystem breakdown (listed above) was used, however FY6 was removed since it only includes Phase F, which does not require manufacturing or test costs.

Fiscal Year Allocation:

- FY1: 21%
- FY2: 30%
- FY3: 30%
- FY4: 16%
- FY5: 3%

1.7.3. Schedule Estimate

The current mission schedule plans a launch date for September 1st, 2028 and an arrival date of September 5th, 2028, which exactly meets the L'Space mission constraint. However, it is likely that the team will be ready for launch well before the current given estimate, and further analogous comparison with similar lunar missions is needed for more accurate progress projection. Phases A and B are excluded from the timeline, and Phase C is assumed to directly start after the conclusion of the L'Space Program. Phases A and B have been excluded from the timeline. Phases C and D utilize data from the Psyche timeline, chosen for its similarities to the Ice Hunter mission in terms of program structure and scope. Phase E utilizes data from SpaceX and estimated timelines for the proposed rover at the Haworth landing site.

The below schedule (including KRB, SRB, and LCR specifics) are based on the NASA Project Management handbook and L'Space team resources. To reiterate, further comparison with similar missions (such as Curiosity and VIPER) is needed to refine these estimates.

Detailed Schedule for Lunar Mission

Phase C: Final Design and Fabrication

- Start Date: October 1, 2024
- End Date: September 30, 2026
- Milestones:
 - Q1 2025: Full-scale assembly of rover instruments and subsystems

- Q3 2025: Systems Integration Review (SIR)
- Q4 2025: Critical Design Review (CDR)

Phase D: System Assembly, Integration, and Test

- Start Date: October 1, 2026
- End Date: August 31, 2028
- Milestones:
 - Q1 2027: Individual instruments and subsystems complete
 - Q2 2027: Testing of individual instruments and subsystems complete
 - Q3 2027: Integrated test complete
 - Q4 2027: Launch Readiness Review (LRR)
 - Q1-Q1 2028: Final Pre-Launch Integration and Checks
 - September 2028: Launch

Phase E: Operations

- Start Date: September 2028
- End Date: November 2028
- Milestones:
 - ~4 days post-launch: Landing on the Moon
 - ~40 days post-landing: Rover in operation

Phase F: Closeout (2029)

- Start Date: December 2028
- End Date: September 2029
- Milestones:
 - Final data analysis and reporting
 - Mission debrief

Key Decision Points and Reviews

KDP-A: Approval to begin Phase A (Concept and Technology Development) - Completed

KDP-B: Approval to begin Phase B (Preliminary Design and Technology Completion) - Completed

KDP-C: Approval to begin Phase C (Final Design and Fabrication) - October 1, 2024

KDP-D: Approval to begin Phase D (System Assembly, Integration, and Test) - October 1, 2026

KDP-E: Approval to begin Phase E (Operations) - September 2028

Standing Review Boards and Lifecycle Reviews

Phase C:

- Systems Integration Review (SIR): Q3 2025
- Critical Design Review (CDR): Q4 2025

Phase D:

- Subsystem Completion Review: Q1 2027
- Testing Completion Review: Q2 2027
- Integrated Test Review: Q3 2027
- Launch Readiness Review (LRR): Q4 2027

1.7.4 Change Control

Changes may be requested by consulting the Project Manager, Deputy Project Manager of Resources, Chief Scientist, or Lead Systems Engineer. After receiving an RFA or ADV, these changes will be tracked in a spreadsheet used for the purpose of tracking changes in documents. After an update, a document will gain a distinguisher marking, such as A, B, C, etc. These will be incremented through the alphabet when a change has occurred. These changes will be communicated throughout the team via the spreadsheet tracking the different versions of the documents, as well as through communication channels throughout the team.

1.8. Conclusion

The overall mission plan is in the preliminary stages. The team faced many challenges throughout the writing of this deliverable including a change in leadership, a lack of communication, and issues distributing the workload evenly among members. An extension was requested and during the extra time, the team first tackled issues with the team structure itself. A meeting was held to ensure each team member had a primary and secondary role. During the MCR, some team members had more than two roles and took on too much of a work load. After the team structure was reconstructed, the team focused on continuing the deliverable. The SRR outlines the complete science goals that drives the mission goals. From the overall mission goals, each subteam defines requirements that will ensure the mission requirements are completed. Along with this, the team goes into detail of cost and schedule estimates and risk analysis. Had the team had more time, more in depth risks analysis and overall detail into subsystem description would be implemented.

For the MDR, the team will communicate who will be completing each specific task of the deliverable and when each team member plans on finishing their part. The team shall communicate clearly if they are feeling stressed or do not have time to complete their part so another team member can lend them aid. The team shall also complete sections of the SRR that were incomplete to prepare for the PDR.

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